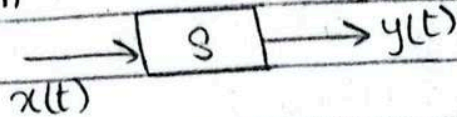
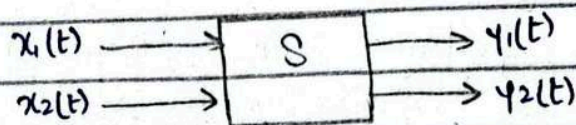


First Week

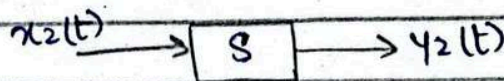
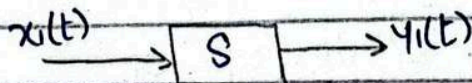
System



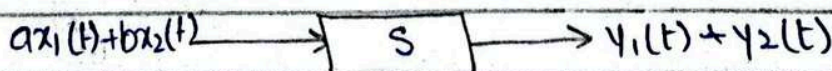
- Single input single output (SISO)
- Multiple input multiple output (MIMO)



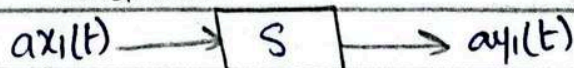
LINEAR System:



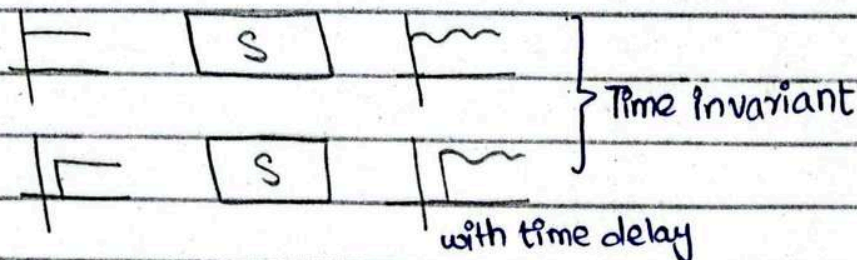
Properties:



↳ addition



↳ homogeneity



* no feedback system — open loop

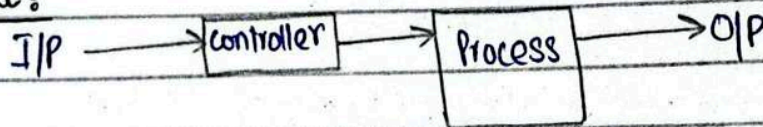
Components of a Control System:

- Input
- Output
- ways to achieve input & output
- control action

Basic Technologies

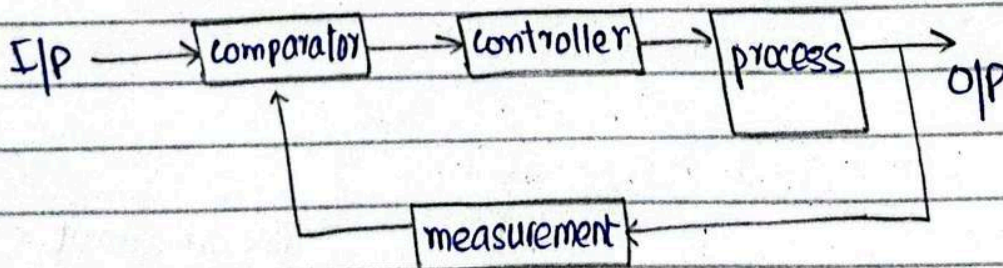
- Plant / Process: portion of the system which is to be controlled or regulated.
- Controller: controls the plant / process
- input
- output
- disturbance

Input:

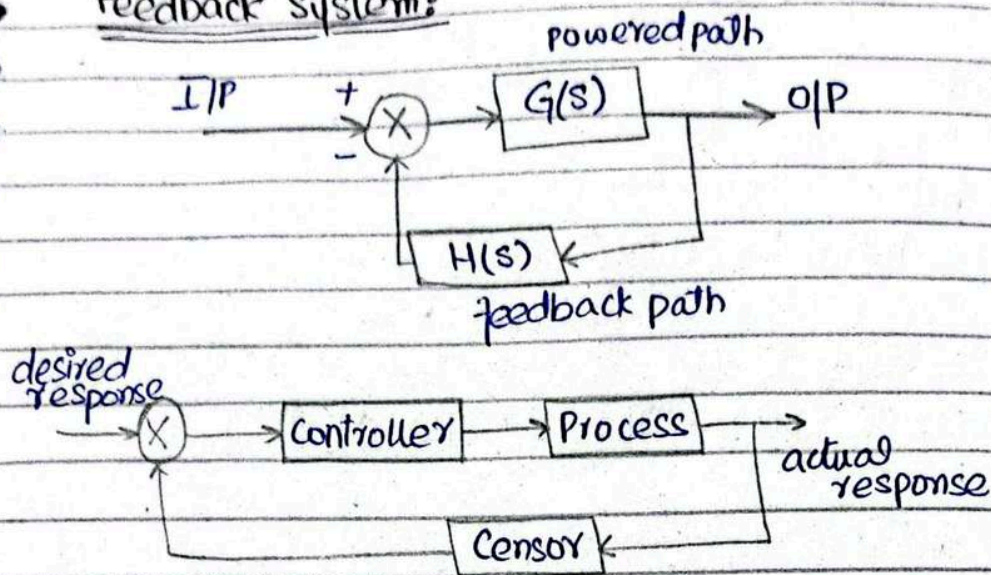


Open loop: O/P is neither measured nor feedback.

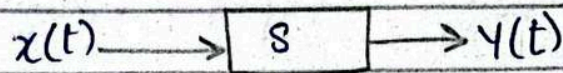
Closed Loop Control System



Feedback System:



Chapter no 2



Transfer function: $T.F = \frac{Y(s)}{X(s)}$

Laplace transform: (covers a broader range of systems)

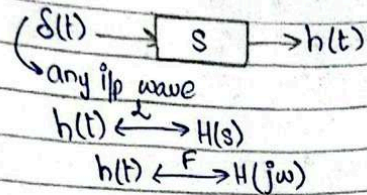
$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

Fourier transform: (for signal analysis)

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

* Fourier is a subset of Laplace transform

Review of Laplace Transforms



$\Rightarrow$ if $s = \sigma + j\omega$

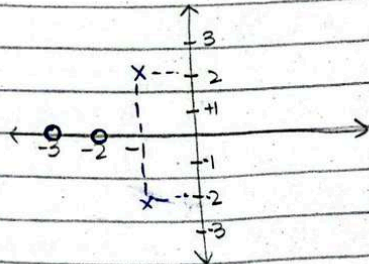
$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} x(t) e^{-\sigma t} e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{Fourier})$$

Example 1 $H_1(s) = \frac{(s+2)(s+3)}{(s+1+2j)(s+1-2j)}$

zeros $\Rightarrow s = -2, s = -3$; poles $\Rightarrow s = -1-2j, s = -1+2j$

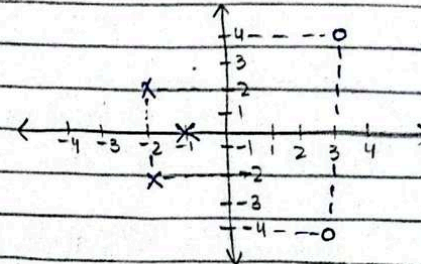
zeros — represented by 'o'
poles — represented by 'x'



Eg #2

$$H_1(s) = \frac{(s-3+4j)(s-3-4j)}{(s+1)(s+2+2j)(s+2-2j)}$$

$$\begin{aligned} s-3+4j=0 & ; s-3-4j=0 & ; s=-1 & ; s+2+2j=0 & , s+2-2j=0 \\ s=3-4j & \quad s=3+4j & & s=-2-2j & \quad s=-2+2j \end{aligned}$$



Impulse related properties

1) $x_1(t) = \delta(t)$

$$X_1(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = e^{-s(0)} = 1$$

2) $x_2(t) = u(t)$

$$\begin{aligned} X_2(s) &= \int_{-\infty}^{\infty} u(t) e^{-st} dt \\ &= \int_0^{\infty} e^{-st} dt = \left. \frac{e^{-st}}{-s} \right|_0^{\infty} = -\frac{1}{s} (e^{-\infty} - e^0) = \frac{1}{s} \end{aligned}$$

3) $x_3(t) = tu(t)$

$$\begin{aligned} X_3(s) &= \int_{-\infty}^{\infty} tu(t) e^{-st} dt = \int_0^{\infty} te^{-st} dt \\ &= \left. te^{-st} \right|_0^{\infty} - \int_0^{\infty} 1 \cdot e^{-st} dt = \frac{1}{s} \left. e^{-st} \right|_0^{\infty} = \frac{1}{s^2} \end{aligned}$$

$$4) x_4(t) = e^{-at} u(t)$$

$$x_4(s) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s+a)t} dt = \frac{e^{-(s+a)t}}{-(s+a)} \Big|_0^{\infty} = \frac{1}{s+a}$$

$$5) x_5(t) = \sin \omega t u(t)$$

$$= \frac{e^{-j\omega t} - e^{j\omega t}}{2j} u(t)$$

$$x_5(s) = \frac{1}{2j} \int_0^{\infty} e^{j\omega t} e^{-st} dt - \frac{1}{2j} \int_0^{\infty} e^{-j\omega t} e^{-st} dt$$

$$= \frac{1}{2j} \int_0^{\infty} e^{-(s-j\omega)t} dt - \frac{1}{2j} \int_0^{\infty} e^{-(s+j\omega)t} dt$$

$$= \frac{1}{2j} \left[\frac{e^{-(s-j\omega)t}}{-(s-j\omega)} \Big|_0^{\infty} - \frac{e^{-(s+j\omega)t}}{-(s+j\omega)} \Big|_0^{\infty} \right]$$

$$x_5(s) = \frac{1}{2j} \left[\frac{1}{s-j\omega} - \frac{1}{s+j\omega} \right] = \frac{1}{2j} \left[\frac{s+j\omega - s+j\omega}{s^2 + \omega^2} \right] = \frac{\omega}{s^2 + \omega^2}$$

Properties of Laplace Transform

1) Linearity:

$$ax_1(t) + bx_2(t) \leftrightarrow ax_1(s) + bx_2(s)$$

2) Time-shifting:

$$x(t-t_0) \leftrightarrow X(s)e^{-st_0}$$

3) Frequency shifting:

$$e^{-at} x(t) \leftrightarrow X(s+a)$$

4) Differentiation property:

$$\frac{dx(t)}{dt} \leftrightarrow sX(s) - x(0)$$

$$\frac{d^2 x(t)}{dt^2} \leftrightarrow s^2 X(s) - sx(0) - x'(0)$$

$$\frac{d^n x(t)}{dt^n} \leftrightarrow s^n X(s) - \sum_{k=1}^n s^{n-k} x^{(k-1)}(0)$$

5) Integration property:

$$\int_0^t x(\tau) d\tau \leftrightarrow \frac{X(s)}{s}$$

③

Initial value theorem

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s)$$

Final value theorem

$$f(\infty) = \lim_{s \rightarrow 0} s(F(s))$$

→ 3 types of analysis

1-Transient 2-steady state 3-stability

$$\rightarrow \frac{d^2 y(t)}{dt^2} + 12 \frac{dy(t)}{dt} + 32 y(t) = 32 x(t)$$

if initial condition are zero

taking Laplace transform

$$s^2 Y(s) + 12s Y(s) + 32 Y(s) = 32 X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{32}{s^2 + 12s + 32}$$

Let say $x(t) = u(t)$, we want to find output

$$X(s) = u(s) = \frac{1}{s}$$

$$Y(s) = \frac{32}{s(s^2 + 12s + 32)} = \frac{32}{s(s+4)(s+8)}$$

$$= \frac{A}{s} + \frac{B}{s+4} + \frac{C}{s+8}$$

$$32 = A(s+4)(s+8) + Bs(s+8) + Cs(s+4)$$

$$s=0$$

$$32 = A(0+4)(0+8), \quad s=-8, \quad s=-4$$

$$A=1; \quad B=-2; \quad C=1$$

$$Y(s) = \frac{1}{s} - \frac{2}{s+4} + \frac{1}{s+8}$$

→ Now taking inverse Laplace transform

$$y(t) = u(t) - 2e^{-4t}u(t) + e^{-8t}u(t)$$

$$= u(t) [1 - 2e^{-4t} + e^{-8t}]$$

$$\rightarrow F(s) = \frac{2}{(s+1)(s+2)^2}$$

$$= \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}$$

$$2 = A(s+2)^2 + B(s+1)(s+2) + C(s+1)$$

$$s=-1, \quad s=-2$$

$$2 = A(-1+2)^2 + B(-1+1)(-1+2) + C(-1+1)$$

$$2 = A(1)^2 \Rightarrow \boxed{A=2}$$

$$2 = A(-2+2)^2 + B(-2+1)(-2+2) + C(-2+1)$$

$$2 = C(-1) \Rightarrow \boxed{C=-2}$$

$$4A + 2B + C = 2$$

$$4(2) + 2B + (-2) = 2$$

$$\boxed{B=-2}$$

$$F(s) = \frac{2}{s+1} - \frac{2}{s+2} - \frac{2}{(s+2)^2}$$

$$f(t) = 2e^{-t}u(t) - 2e^{-2t}u(t) - 2e^{-2t}t u(t)$$

$$= 2u(t) [e^{-t} - e^{-2t} - te^{-2t}]$$

$$\rightarrow F(s) = \frac{3}{s(s^2 + 2s + 5)}$$

$$= \frac{A}{s} + \frac{Bs+C}{s^2 + 2s + 5}$$

$$3 = A(s^2 + 2s + 5) + Bs + C(s)$$

$$s=0$$

$$\boxed{A=3}$$

Now,

$$A+B=0; \quad 2A+C=0$$

$$B = -\frac{3}{5} ; C = -\frac{6}{5}$$

$$F(s) = \frac{3/5}{s} - \frac{3/5s + 6/5}{s^2 + 2s + 5}$$

$$= \frac{3/5}{s} - \frac{3}{5} \frac{s+2}{s^2 + 2s + 5} = \frac{3/5}{s} - \frac{3}{5} \frac{s+2}{s^2 + 2s + 1 + 4}$$

$$= \frac{3/5}{s} - \frac{3}{5} \frac{s+2}{(s+1)^2 + (2)^2}$$

$$= \frac{3/5}{s} - \frac{3}{5} \left[\frac{s+1}{(s+1)^2 + 2^2} + \frac{1}{(s+1)^2 + 2^2} \right]$$

$$F(s) = \frac{3/5}{s} - \frac{3}{5} \left[\frac{s+1}{(s+1)^2 + 2^2} + \frac{1}{2} \frac{2}{(s+1)^2 + 2^2} \right]$$

$$f(t) = \frac{3}{5} u(t) - \frac{3}{5} \left[e^{-t} \cos 2t + \frac{1}{2} e^{-t} \sin 2t \right] u(t)$$

$$= \left\{ \frac{3}{5} - \frac{3}{5} e^{-t} \left[\cos 2t + \frac{1}{2} \sin 2t \right] \right\} u(t)$$

$$= 0.6 - 0.67 e^{-t} \left[\frac{1}{\sqrt{1+(1/2)^2}} \cos 2t + \frac{1/2}{\sqrt{1+(1/2)^2}} \sin 2t \right]$$

$$\therefore \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$= 0.6 u(t) - 0.67 e^{-t} \cos(\alpha t - \phi) u(t)$$

$$\phi = \tan^{-1} \frac{1}{2}$$

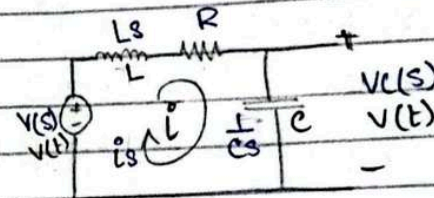
$$G(s) = \frac{2s+1}{s^2+2s+6}$$

$$(s^2+2s+6) Y(s) = (2s+1) X(s)$$

$$s^2 Y(s) + 2s Y(s) + 6 Y(s) = 2s X(s) + X(s)$$

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 6 y(t) = 2 \frac{dx(t)}{dt} + x(t)$$

Transfer function of Electrical Networks:



$$\rightarrow i = C \frac{dV_c}{dt}$$

$$I(s) = Cs V_c(s)$$

$$\rightarrow V_c = L \frac{di}{dt}$$

$$V_c(s) = Ls I(s)$$

$$\text{Find } \frac{V_c(s)}{V(s)}$$

$$-V(t) + L \frac{di}{dt} + R i + V_c(t) = 0$$

$$-V(t) + LC \frac{d^2 V_c(t)}{dt^2} + RC \frac{dV_c}{dt} + V_c(t) = 0$$

taking Laplace transform

$$LCs^2 V_c(s) + RCs V_c(s) + V_c(s) + V(s) = 0$$

$$\begin{aligned} \frac{V_c(s)}{V(s)} &= \frac{1}{LCs^2 + RCs + 1} \\ &= \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}} \end{aligned}$$

→ In terms of 's'

$$-V(s) + LsI(s) + RI(s) + I(s)\frac{1}{Cs} = 0$$

$$(Ls + R + \frac{1}{Cs})I(s) = V(s)$$

$$(Ls + R + \frac{1}{Cs})(Cs)(V_c(s)) = V(s)$$

$$(Lcs^2 + Rcs + 1)(V_c(s)) = V(s)$$

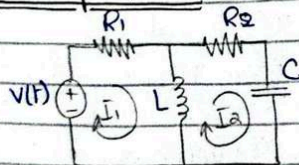
$$\frac{V_c(s)}{V(s)} = \frac{1}{Lcs^2 + Rcs + 1}$$

→ 3rd Method

Apply VDR:

$$V_c(s) = V(s) \frac{1/Cs}{1/Cs + R + Ls}$$

Multiple Loop Circuit



$$-V(s) + R_1 I_1(s) + Ls(I_1(s) - I_2(s)) = 0$$

$$R_2 I_2(s) + \frac{1}{Cs} I_2(s) + Ls(I_2(s) - I_1(s)) = 0$$

$$(Ls + R_1)I_1(s) - LsI_2(s) = V(s)$$

$$-LsI_1(s) + (\frac{1}{Cs} + (Ls + R_2))I_2(s) = 0$$

$$\begin{bmatrix} Ls + R_1 & -Ls \\ -Ls & \frac{1}{Cs} + Ls + R_2 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V(s) \\ 0 \end{bmatrix}$$

$$I_1(s) = \frac{\begin{vmatrix} V(s) & -Ls \\ 0 & R_2 + Ls + \frac{1}{Cs} \end{vmatrix}}{\Delta}$$

$$I_2(s) = \frac{\begin{vmatrix} Ls + R_1 & V(s) \\ -Ls & 0 \end{vmatrix}}{\Delta}$$

$$\Delta = \begin{vmatrix} Ls + R_1 & -Ls \\ -Ls & R_2 + Ls + \frac{1}{Cs} \end{vmatrix}$$

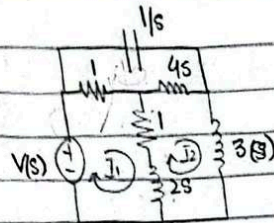
$$= (Ls + R_1)(R_2 + \frac{1}{Cs} + Ls) - L^2 s^2$$

Shortcut Method:

$$\left[\begin{array}{c} \text{sum of imp} \\ \text{around Mesh 1} \end{array} \right] I_1(s) - \left[\begin{array}{c} \text{sum of imp} \\ \text{common in} \\ \text{1 and 2} \end{array} \right] I_2(s) = \left[\begin{array}{c} \text{sum of} \\ \text{voltage in 1} \end{array} \right]$$

$$\left[\begin{array}{c} \text{sum of imp} \\ \text{around Mesh 2} \end{array} \right] I_2(s) - \left[\begin{array}{c} \text{sum of imp} \\ \text{common in} \\ \text{1 and 2} \end{array} \right] I_1(s) = \left[\begin{array}{c} \text{sum of} \\ \text{voltage in 2} \end{array} \right]$$

(4)

Find $\frac{I_1(s)}{V(s)}$

Mesh 1:

$$(2+2s)(I_1(s)) - (1+2s)(I_2(s)) - (1)(I_3(s)) = V_s \quad \text{--- (1)}$$

Mesh 2:

$$(1+9s)(I_2(s)) - (1+2s)(I_1(s)) - 4s(I_3(s)) = 0 \quad \text{--- (2)}$$

Mesh 3:

$$(1+4s+\frac{1}{s})(I_3(s)) - (1)(I_1(s)) - 4s(I_2(s)) = 0 \quad \text{--- (3)}$$

$$\begin{bmatrix} 2+2s & -(1+2s) & -1 \\ -(1+2s) & 9s+1 & -4s \\ -1 & -4s & \frac{1}{s}+4s+1 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \\ 0 \end{bmatrix}$$

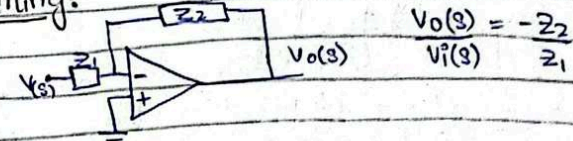
$$I_1(s) = \frac{\begin{vmatrix} V_s & -(1+2s) & -1 \\ 0 & (9s+1) & -4s \\ 0 & -4s & \frac{1}{s}+4s+1 \end{vmatrix}}{\Delta}$$

 Δ

$$\Delta = \begin{vmatrix} 2+2s & -(1+2s) & -1 \\ -(1+2s) & 9s+1 & -4s \\ -1 & -4s & \frac{1}{s}+4s+1 \end{vmatrix}$$

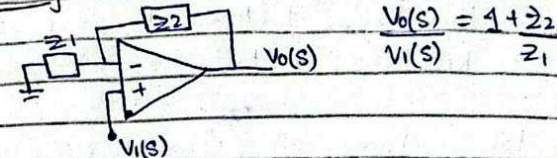
Transfer function of Active ccts / OP-AMP based ccts

Inverting:



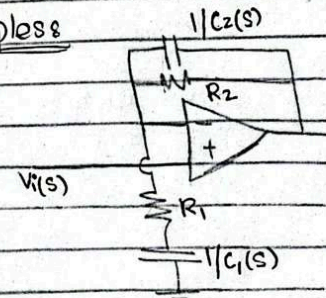
$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2}{Z_1}$$

Non-Inverting:



$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{Z_2}{Z_1}$$

Examples:



$$Z_2 = R_2 \parallel \frac{1}{C_2(s)} = \frac{R_2}{R_2 C_2(s) + 1}$$

$$= \frac{R_2}{1 + R_2 C_2(s)} \quad \text{--- (1)}$$

$$Z_1 = R_1 + \frac{1}{C_1(s)} = \frac{1 + R_1 C_1(s)}{C_1(s)} \quad \text{--- (2)}$$

$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{R_2 \left(1 + R_2 C_2(s) \right)}{1 + R_1 C_1(s)} \cdot \frac{C_1(s)}{C_1(s)}$$

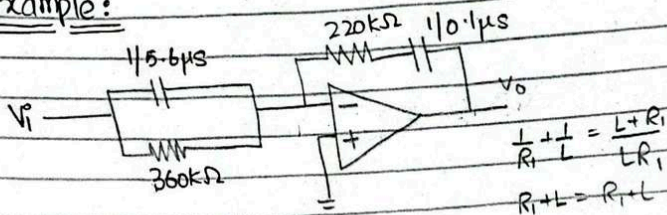
$$= 1 + \frac{R_2 C_1(s)}{(1 + R_2 C_2(s))(1 + R_1 C_1(s))}$$

$$= \frac{1 + (R_1 C_1(s)(1 + R_2 C_2(s)) + R_2 C_1(s))}{(1 + R_1 C_1(s))(1 + R_2 C_2(s))}$$

$$= \frac{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1)s + 1}{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2)s + 1}$$

∴ Poles = Power of s in denominator
 ∴ Zeros = Power of s in numerator

Example:



$$Z_2 = \frac{220k + 10^7}{s}, \quad Z_1 = \frac{360k}{1 + (360k)(5.6\mu)s}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{220k + 10^7/s}{360k / (1 + (360k)(5.6\mu)s)}$$

$$= - \left[\frac{220k s + 10^7}{s} \times \left(\frac{(1 + 2.016)s}{360k} \right) \right]$$

$$= - \left(\frac{1.232s^2 + 56.61s + 27.77}{s} \right)$$

$$= - \left[1.232s + \frac{27.77}{s} + 56.61 \right]$$

Transfer Function of Mechanical Systems

→ Translational Mechanical System → Mass, Spring, Damper
 → Force, Displacement

Mass:

→ stores kinetic energy
 → rigid bodies

∴ Number of equations

depends on the point of displacement.

$$f(t) \propto a$$

$$f(t) = Ma \Rightarrow \boxed{f(t) = M \frac{d^2 y(t)}{dt^2}}$$

2- Spring:

→ stores potential energy

$$f(t) = k(y_1(t) - y_2(t))$$

$$f(t) \propto y(t)$$

$$\boxed{f(t) = k y(t)}$$

3- Dampers:

→ Frictional element

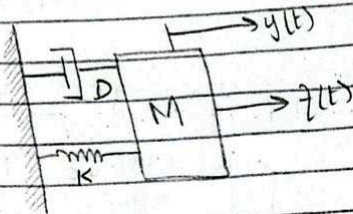
$$\tau(t) \propto v(t)$$

$$\tau(t) \propto \frac{dy(t)}{dt}$$

$$\tau(t) = \tau_v \frac{dy(t)}{dt}$$

→ τ_v = viscous friction coefficient

Ideal Spring Mass Damper System



FBD at y_1 :

$$\tau(t) = M \frac{d^2 y}{dt^2} + \tau_v \frac{dy}{dt} + k y(t)$$

Taking Laplace transform

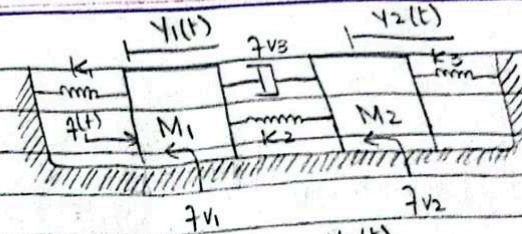
$$F(s) = M s^2 Y(s) + \tau_v s Y(s) + k Y(s)$$

$$Y(s) [M s^2 + \tau_v s + k] = F(s)$$

$$\frac{Y(s)}{F(s)} = \frac{1}{M s^2 + \tau_v s + k}$$

$$M=3, \tau_v=2, k=1$$

$$\frac{Y(s)}{F(s)} = \frac{1}{3s^2 + 2s + 1}$$



FBD at y_1 :

$$\tau(t) = M_1 \frac{d^2 y_1}{dt^2} + \tau_{v1} \frac{dy_1}{dt} + \tau_{v3} \left[\frac{dy_1}{dt} - \frac{dy_2}{dt} \right] + k_1 y_1 + k_2 (y_1 - y_2)$$

$$\tau(t) = M_1 \frac{d^2 y_1}{dt^2} + \tau_{v1} \frac{dy_1}{dt} + \tau_{v3} \left[\frac{dy_1}{dt} - \frac{dy_2}{dt} \right] + k_1 y_1 + k_2 (y_1 - y_2)$$

FBD at y_2 :

$$k_2 (y_2 - y_1) \leftarrow \frac{M_2 d^2 y_2}{dt^2} \leftarrow \tau_{v2} \frac{dy_2}{dt} \leftarrow k_3 y_2$$

$$M_2 \frac{d^2 y_2}{dt^2} + \tau_{v2} \frac{dy_2}{dt} + k_3 y_2 + k_2 (y_2 - y_1) + \tau_{v3} \left[\frac{dy_2}{dt} - \frac{dy_1}{dt} \right] = 0$$

$$\Rightarrow [M_1 s^2 (\tau_{v1} + \tau_{v3}) s + (k_1 + k_2)] Y_1(s) - (\tau_{v3} s + k_2) Y_2(s) = F(s)$$

$$\Rightarrow -(\tau_{v3} s + k_2) Y_1(s) + [M_2 s^2 + (\tau_{v2} - \tau_{v3}) s + k_2 + k_3] Y_2(s) = 0$$

CRAMER'S RULE:

$$\begin{bmatrix} M_1 s^2 + (\tau_{v1} + \tau_{v3}) s + k_1 + k_2 & -\tau_{v3} s + k_2 \\ -(\tau_{v3} s + k_2) & M_2 s^2 + (\tau_{v2} - \tau_{v3}) s + k_2 + k_3 \end{bmatrix} \begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

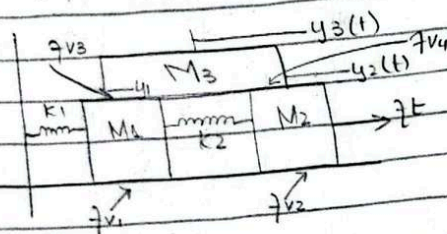
* Technique:

$$\left[\begin{array}{c} \text{sum of imp at} \\ \text{motion 1} \end{array} \right] Y_1(s) - \left[\begin{array}{c} \text{sum of imp} \\ \text{common in land 2} \end{array} \right] Y_2(s) = \left[\begin{array}{c} \text{sum of} \\ \text{applied forces} \\ \text{at } Y_1 \end{array} \right]$$

$$\left[\begin{array}{c} \text{sum of imp at} \\ \text{motion 2} \end{array} \right] Y_2(s) - \left[\begin{array}{c} \text{sum of imp} \\ \text{common} \end{array} \right] Y_1(s) = \left[\begin{array}{c} \text{sum of} \\ \text{applied forces} \\ \text{at } Y_2 \end{array} \right]$$

⑥

Example:



Equation by using technique.

For M1:

$$(M_1 s^2 + (f_{v1} + f_{v3})s + k_1 + k_2) Y_1(s) - k_2 Y_2(s) - (f_{v3} s) Y_3(s) = 0$$

For M2:

$$(M_2 s^2 + (f_{v2} + f_{v4})s + k_2) Y_2(s) - k_2 Y_1(s) - (f_{v4} s) Y_3(s) = F(s)$$

For M3:

$$(M_3 s^2 + (f_{v3} + f_{v4})s) Y_3(s) - (f_{v3} s) Y_1(s) - (f_{v4} s) Y_2(s) = 0$$

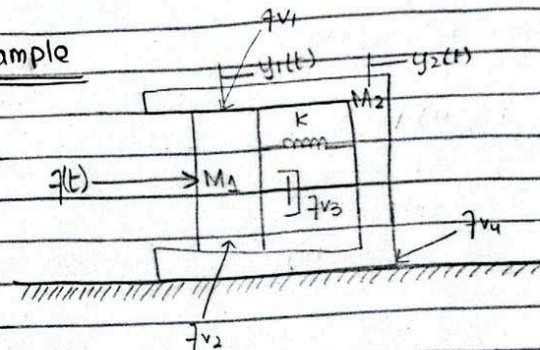
Equation using cramer's rule:

$$\begin{bmatrix} M_1 s^2 + (f_{v1} + f_{v3})s + k_1 + k_2 & -k_2 & -f_{v3} s \\ M_2 s^2 + (f_{v2} + f_{v4})s + k_2 & -f_{v4} s & 0 \\ -k_2 & -f_{v4} s & M_3 s^2 + (f_{v3} + f_{v4})s \end{bmatrix} \begin{bmatrix} Y_1(s) \\ Y_2(s) \\ Y_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ F(s) \\ 0 \end{bmatrix}$$

Remember!

Matrix mein M, k, f ki value put karni hai

Example



Take $f_{v1} = f_{v2} = f_{v3} = f_{v4} = 1 \text{ N/s/m}$

$M_1 = M_2 = 1 \text{ kg}$

$k = 1 \text{ N/m}$

Find $\frac{Y_2(s)}{F(s)}$

For M1:

$$(M_1 s^2) Y_1(s) - (f_{v1} + f_{v2} + f_{v3} + k) Y_2(s) = F(s)$$

$$\rightarrow (M_1 s^2 + (f_{v1} + f_{v2} + f_{v3})s + k) Y_1(s) - (f_{v1} + f_{v2} + f_{v3} + k) Y_2(s) = F(s)$$

For M_2 :

$$(M_2 s^2 + (7v_1 + 7v_2 + 7v_3 + 7v) s + k) - [(7v_1 + 7v_2 + 7v_3) s + k] y_1(s) = 0$$

Cramer's Rule

$$\begin{bmatrix} s^2 + 3s + 1 & -(3s + 1) \\ -(3s + 1) & s^2 + 4s + 1 \end{bmatrix} \begin{bmatrix} y_1(s) \\ y_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$y_2(s) = \frac{\begin{vmatrix} s^2 + 3s + 1 & F(s) \\ -(3s + 1) & 0 \end{vmatrix}}{\Delta}$$

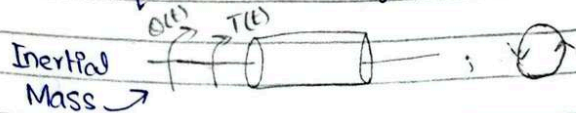
$$\Delta = s(s^3 + 7s^2 + 5s + 1)$$

$$y_2(s) = \frac{(3s + 1) F(s)}{\Delta}$$

$$\frac{y_2(s)}{F(s)} = \frac{3s + 1}{s(s^3 + 7s^2 + 5s + 1)}$$

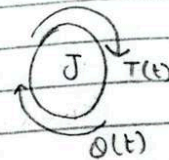
Rotational Mechanical System

→ Torque $T(t)$ → Angular displacement $\theta(t)$

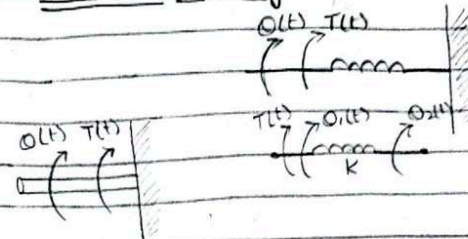


$$T(t) \propto \frac{d^2 \theta(t)}{dt^2} ; T(t) = J \frac{d^2 \theta(t)}{dt^2}$$

J = Moment of Inertia



→ Torsional Spring:

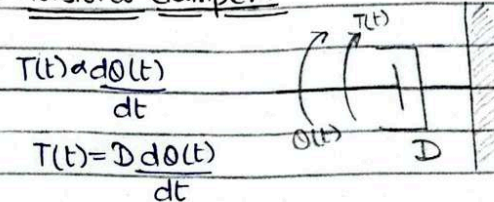


$$T(t) \propto \theta(t)$$

$$T(t) = K \theta(t)$$

$$T(t) = K (\theta_1(t) - \theta_2(t))$$

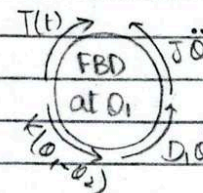
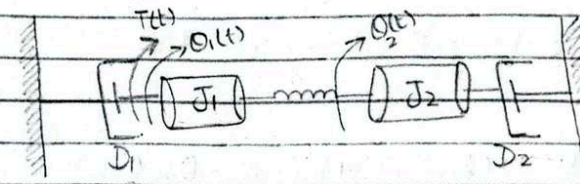
→ Torsional damper:



$$T(t) \propto \frac{d\theta(t)}{dt}$$

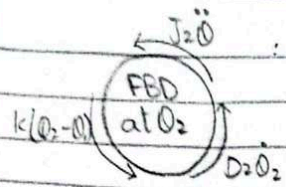
$$T(t) = D \frac{d\theta(t)}{dt}$$

Example



$$J_1 \frac{d^2 \theta_1}{dt^2} + D \frac{d\theta_1}{dt} + k(\theta_1 - \theta_2) = T(t)$$

$$\hookrightarrow (J_1 s^2 + D s + k) \theta_1(s) - k \theta_2(s) = T(s)$$



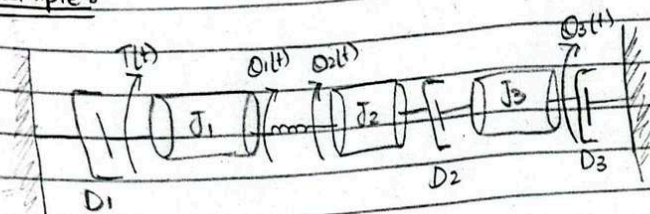
$$J_2 \frac{d^2 O_2}{dt^2} + D_2 \frac{dO_2}{dt} + k(O_2 - O_1) = 0$$

$$-k O_1(s) + (J_2 s^2 + D_2 s + k) O_2(s) = 0$$

Cramer's Rule:

$$\begin{bmatrix} J_1 s^2 + D_1 s + k & -k \\ -k & J_2 s^2 + D_2 s + k \end{bmatrix} \begin{bmatrix} O_1 \\ O_2 \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \end{bmatrix}$$

Example:



For O_1 :

$$(J_1 s^2 + D_1 s + k_1) O_1(s) - k O_2(s) = T(s)$$

For O_2 :

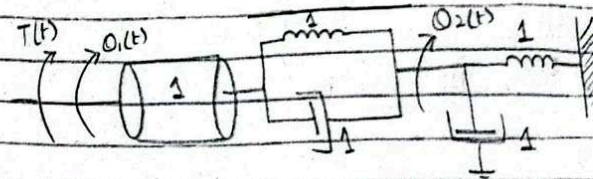
$$-k O_1(s) + (J_2 s^2 + D_2 s + k) O_2(s) - D_2 s O_3(s) = 0$$

For O_3 :

$$-D_2 s O_2(s) + (J_3 s^2 + (D_2 + D_3) s) O_3(s) = 0$$

Cramer's Rule:

$$\begin{bmatrix} J_1 s^2 + D_1 s + k & -k & 0 \\ -k & J_2 s^2 + D_2 s + k & -D_2 s \\ 0 & -D_2 s & J_3 s^2 + (D_2 + D_3) s \end{bmatrix} \begin{bmatrix} O_1(s) \\ O_2(s) \\ O_3(s) \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \\ 0 \end{bmatrix}$$



Find $\frac{O_2(s)}{T(s)}$

$$(s^2 + s + 1) O_1(s) - (s + 1) O_2(s) = T(s)$$

$$-(s + 1) O_1(s) + (2s + 2) O_2(s) = 0$$

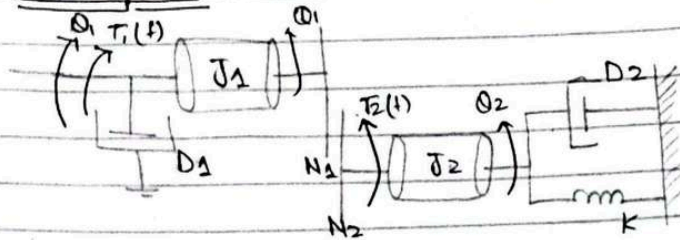
Cramer's Rule:

$$\begin{bmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & 2s + 2 \end{bmatrix} \begin{bmatrix} O_1 \\ O_2 \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \end{bmatrix}$$

$$O_2(s) = \frac{\begin{vmatrix} s^2 + s + 1 & T(s) \\ -(s + 1) & 0 \end{vmatrix}}{\Delta}$$

$$\Delta = \begin{vmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & 2s + 2 \end{vmatrix}$$

Example:



i) $\frac{\Theta_2(s)}{T_2(s)} = ?$

→ at Θ_2 :

$$J_{eq} = J_2 + J_1 \left(\frac{N_2}{N_1} \right)^2$$

$$D_{eq} = D_2 + D_1 \left(\frac{N_2}{N_1} \right)^2$$

$$\frac{\Theta_2(s)}{T_2(s)} = \frac{k_{eq}}{J_{eq}s^2 + D_{eq}s + k_{eq}}$$

ii) $\frac{\Theta_1(s)}{T_1(s)} = ?$

$$(J_{eq}s^2 + D_{eq}s + k_{eq}) \Theta_1(s) = T_1(s)$$

$$J_{eq} = J_1 + J_2 \left(\frac{N_1}{N_2} \right)^2$$

$$D_{eq} = D_1 + D_2 \left(\frac{N_1}{N_2} \right)^2$$

$$k_{eq} = k \left(\frac{N_1}{N_2} \right)^2$$

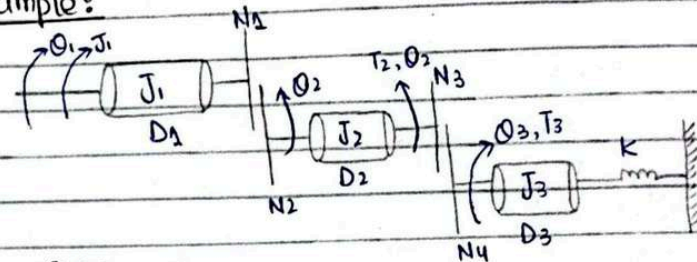
iii) $\frac{\Theta_2(s)}{T_1(s)} = ?$

$$(J_{eq}s^2 + D_{eq}s + k_{eq}) \Theta_2(s) = T_2(s)$$

$$(J_{eq}s^2 + D_{eq}s + k_{eq}) \Theta_2(s) = \frac{N_2}{N_1} T_1(s)$$

$$\frac{\Theta_2(s)}{T_1(s)} = \frac{(N_2/N_1)}{J_{eq}s^2 + D_{eq}s + k_{eq}}$$

Example:



$\frac{\Theta_3(s)}{T_3(s)} = ?$

$$J_{eq} = J_3 + J_2 \left(\frac{N_4}{N_3} \right)^2 + J_1 \left(\frac{N_4}{N_3} \right)^2 \left(\frac{N_2}{N_1} \right)^2$$

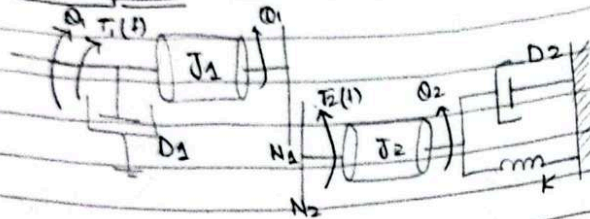
$$D_{eq} = D_3 + D_2 \left(\frac{N_4}{N_3} \right)^2 + D_1 \left(\frac{N_4}{N_3} \right)^2 \left(\frac{N_2}{N_1} \right)^2$$

$$k_{eq} = k$$

Transfer Function of ElectroMechanical System
(DC Servomotor)

1. Servomotor is with feedback loop.

Example:



i) $\frac{\Theta_2(s)}{T_2(s)} = ?$

→ at Θ_2 :

$$J_{eq} = J_2 + J_1 \left(\frac{N_2}{N_1} \right)^2$$

$$D_{eq} = D_2 + D_1 \left(\frac{N_2}{N_1} \right)^2$$

$$K_{eq} = k$$

$$\frac{\Theta_2(s)}{T_2(s)} = \frac{1}{J_{eq}s^2 + D_{eq}s + K_{eq}}$$

ii) $\frac{\Theta_1(s)}{T_1(s)} = ?$

$$(J_{eq}s^2 + D_{eq}s + K_{eq}) \Theta_1(s) = T_1(s)$$

$$J_{eq} = J_1 + J_2 \left(\frac{N_1}{N_2} \right)^2$$

$$D_{eq} = D_1 + D_2 \left(\frac{N_1}{N_2} \right)^2$$

$$K_{eq} = k \left(\frac{N_1}{N_2} \right)^2$$

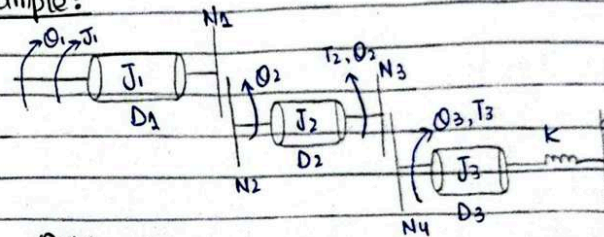
iii) $\frac{\Theta_2(s)}{T_1(s)} = ?$

$$(J_{eq}s^2 + D_{eq}s + K_{eq}) \Theta_2(s) = T_2(s)$$

$$(J_{eq}s^2 + D_{eq}s + K_{eq}) \Theta_2(s) = \frac{N_2}{N_1} T_1(s)$$

$$\frac{\Theta_2(s)}{T_1(s)} = \frac{(N_2/N_1)}{J_{eq}s^2 + D_{eq}s + K_{eq}}$$

Example:



$\frac{\Theta_3(s)}{T_3(s)} = ?$

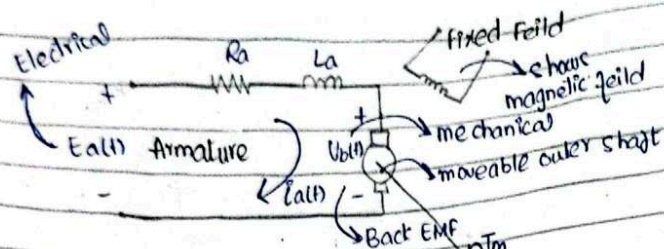
$$J_{eq} = J_3 + J_2 \left(\frac{N_4}{N_3} \right)^2 + J_1 \left(\frac{N_4}{N_3} \right)^2 \left(\frac{N_2}{N_1} \right)^2$$

$$D_{eq} = D_3 + D_2 \left(\frac{N_4}{N_3} \right)^2 + D_1 \left(\frac{N_4}{N_3} \right)^2 \left(\frac{N_2}{N_1} \right)^2$$

$$K_{eq} = k$$

Transfer Function of ElectroMechanical System
(DC Servomotor)

1. Servomotor is with feedback loop.



$$V_b(t) \propto \frac{d\theta_m}{dt}$$

back emf constant

$$V_b(t) = k_b \frac{d\theta_m}{dt}$$

$$\Rightarrow V_b(s) = k_b s \theta_m(s) \quad \text{--- (i)}$$

Transfer Function:

$$\frac{\theta_m(s)}{E_a(s)}$$

$$T_m(t) \propto i_a(t)$$

$$T_m(t) = k_t i_a(t) \Rightarrow T_m(s) = k_t I_a(s) \quad \text{--- (ii)}$$

motor torque constant

$$-E_a(t) + R_a i_a(t) + L_a \frac{di_a}{dt} + V_b(t) = 0$$

$$-E_a(s) + R_a I_a(s) + L_a s I_a(s) + V_b(s) = 0$$

$$(R_a + L_a s) I_a(s) + k_b s \theta_m(s) = E_a(s) \quad \text{--- (iii)}$$

eq (i)

$$\rightarrow \text{(ii)} \quad I_a(s) = \frac{T_m(s)}{k_t}$$

$$\rightarrow \text{(iii)} \quad (R_a + L_a s) \frac{T_m(s)}{k_t} + k_b s \theta_m(s) = E_a(s) \quad \text{--- (iv)}$$

$$J_m \frac{d^2 \theta_m}{dt^2} + D_m \frac{d\theta_m}{dt} = T_m(t)$$

$$(J_m s^2 + D_m) \theta_m(s) = T_m(s) \quad \text{--- (v)}$$

$$\rightarrow \text{(iv)} \quad (R_a + L_a s) \frac{(J_m s^2 + D_m) \theta_m(s) + k_b s \theta_m(s)}{k_t} = E_a(s)$$

$$\left[\frac{(R_a + L_a s)(J_m s^2 + D_m) + k_b(s)}{k_t} \right] \theta_m(s) = E_a(s)$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{k_t}{(R_a + L_a s)(J_m s^2 + D_m) + k_b(s) k_t}$$

$$L_a \rightarrow 0$$

$$= \frac{k_t / R_a J_m}{s^2 + (D_m / J_m) s + (k_b k_t / R_a J_m)}$$

$$= \frac{k_t / R_a J_m}{s(s + \frac{1}{J_m}(D_m + \frac{k_b k_t}{R_a}))} = \frac{k}{s(s + a)}$$

$$= \frac{k}{s(s + a)} \quad \therefore \alpha = \frac{1}{J_m} (D_m + \frac{k_b k_t}{R_a})$$

$$\text{Poles: } s = 0, s = -\alpha$$

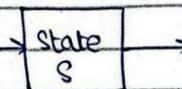
$$\therefore K = \frac{k_t}{R_a J_m}$$

Chapter no 03

State Space Representation

for non-linear time equation

→ input, output
→ state variables



$x_1(t), x_2(t), x_3(t), \dots$

$\dot{X} = AX + BU$ (state vector)

$Y = CX + DU$

(input vector)

$A \rightarrow$ state matrix

$B \rightarrow$ input matrix, $C \rightarrow$ output matrix

$D \rightarrow$ Direct transmission matrix

- How many state variables?
- How to assign state variables?
- Write state equations * 1st order } state space model
- Write output equations

$$\begin{matrix} n \times 1 & n \times n & n \times m \\ X & = & AX + BU \\ & \downarrow & \downarrow \\ & n \times 1 & m \times 1 \end{matrix}$$

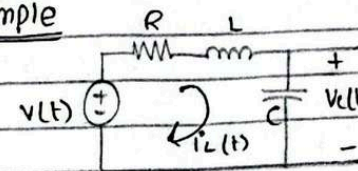
$\therefore P =$ output variables
 $n =$ state variables
 $m =$ input variables

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{matrix} p \times 1 & p \times n & p \times m \\ Y & = & CX + DU \\ & \downarrow & \downarrow \\ & n \times 1 & m \times 1 \end{matrix}$$

$\therefore$ no of state variables = no. of state equations

Example



$\Rightarrow$ *energy storing variables = 2

$\Rightarrow$ state variables = 2

Let,

$$x_1(t) = v_o(t) = v_L(t) \quad \text{--- ①}$$

$$x_2(t) = i_L(t)$$

$$i_L(t) = C \frac{dv_L(t)}{dt}$$

$$\Rightarrow x_2 = C \frac{dx_1}{dt} \Rightarrow \frac{dx_1}{dt} = \frac{1}{C} x_2 \quad \text{--- ②}$$

$$-v(t) + R i_L(t) + L \frac{di_L}{dt} + v_L(t) = 0$$

$$\frac{di_L}{dt} = -\frac{1}{L} v_L(t) - \frac{R}{L} i_L(t) + \frac{1}{L} v(t)$$

$$\frac{dx_2}{dt} = -\frac{1}{L} x_1 - \frac{R}{L} x_2 + \frac{1}{L} v(t) \quad \text{--- ③}$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \end{bmatrix} = \begin{bmatrix} 0 & 1/C \\ -1/L & -R/L \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} v(t)$$

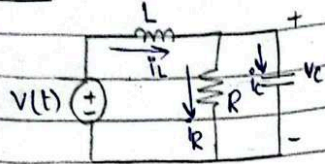
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Let, $v_R(t) = v_o$

$$v_R(t) = R x_2$$

$$\begin{bmatrix} 0 & R \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Example



Let,

$$V_c = x_1 \quad ; \quad i_L = x_2$$

$$\therefore i_L = i_R + i_C$$

$$i_C = C \frac{dV_c}{dt} \Rightarrow i_C = C \frac{dx_1}{dt} \Rightarrow \frac{dx_1}{dt} = \frac{1}{C} i_C = \frac{1}{C} (i_L - i_R)$$

$$= \frac{1}{C} \left(i_L - \frac{V_c}{R} \right)$$

$$= \frac{1}{C} x_2 - \frac{1}{RC} x_1 \Rightarrow \frac{dx_1}{dt} = -\frac{1}{RC} x_1 + \frac{1}{C} x_2 \quad \text{--- (i)}$$

$$V_L = L \frac{di_L}{dt} \Rightarrow \frac{dx_2}{dt} = \frac{1}{L} V_L \quad \text{--- (ii)}$$

$$-V(t) + V_L + V_R = 0$$

$$V_L = V(t) - V_c$$

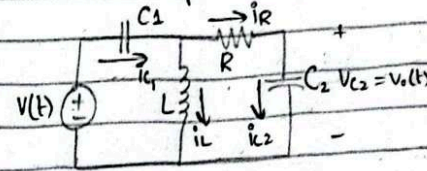
$$\frac{dx_2}{dt} = \frac{1}{L} (-V_c + V(t))$$

$$\frac{dx_2}{dt} = -\frac{1}{L} x_1 + \frac{1}{L} V(t) \quad \text{--- (iii)}$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \end{bmatrix} = \begin{bmatrix} -1/RC & 1/C \\ -1/L & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} V(t)$$

$$y = \begin{bmatrix} -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + V(t)$$

Obtain state space model. Take v_{c2} as output



Let,

$$V_{c2} = x_3 = v_o(t) \quad x_1, x_2, x_3 \rightarrow \text{State Variables}$$

$$V_{c1} = x_1$$

$$i_L = x_2 \Rightarrow \frac{dx_2}{dt} = C \frac{dV_{c2}}{dt}$$

$$i_{C1} = C_1 \frac{dV_{c1}}{dt} \Rightarrow \frac{1}{C_1} i_{C1} = \frac{dV_{c1}}{dt} \quad \text{--- (i)}$$

$$i_{C2} = C_2 \frac{dV_{c2}}{dt} \Rightarrow \frac{1}{C_2} i_{C2} = \frac{dV_{c2}}{dt} \quad \text{--- (ii)}$$

$$V_L = L \frac{di_L}{dt} \Rightarrow \frac{1}{L} V_L = \frac{di_L}{dt} \quad \text{--- (iii)}$$

Replace i_{C2} and V_L

$$-V(t) + V_{C1}(t) + V_L(t) = 0$$

$$\text{--- (iii)} \quad -V(t) + V_{C1} + V_L = 0$$

$$V_L = V(t) - V_{C1}$$

--- (iii)

$$\frac{di_L}{dt} = \frac{1}{L} (V(t) - V_{C1}) \Rightarrow \frac{dx_2}{dt} = \frac{1}{L} (V(t) - x_1)$$

$$i_{C1} = i_R + i_L$$

$$i_R = \frac{V_L - V_{C2}}{R}$$

$$\rightarrow (v) \quad i_1 = i_L + \left(\frac{V_L - V_{C2}}{R} \right)$$

$$i_1 = i_L + \frac{1}{R} \left[\underbrace{-V_{C1} + v(t)}_{V_L} - V_{C2} \right]$$

$$\frac{dV_{C1}}{dt} = \left[\frac{1}{C_1} \left(i_L + \frac{1}{R} (-V_{C1} + v(t) - V_{C2}) \right) \right]$$

$$\frac{dx_1}{dt} = \frac{1}{C_1} x_2 - \frac{1}{RC_1} x_1 - \frac{1}{RC_1} x_3 + \frac{1}{RC_1} v(t)$$

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \end{bmatrix} = \begin{bmatrix} -1/RC_1 & 1/C_1 & -1/RC_1 \\ 1/L & 0 & 0 \\ -1/RC_2 & 0 & -1/RC_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1/RC_1 \\ 1/L \\ 1/RC_2 \end{bmatrix} v(t)$$

$\rightarrow V_{C2}$

$$y(t) = V_{C2}$$

$$y = x_3$$

$$[y] = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\rightarrow (v) \quad i_{c1} = i_L + \left(\frac{V_L - V_{C2}}{R} \right)$$

$$i_{c1} = i_L + \frac{1}{R} \left[\underbrace{-V_{C1} + v(t)}_{V_L} - V_{C2} \right]$$

$$\frac{dV_{C1}}{dt} = \left[\frac{1}{C_1} \left(i_L + \frac{1}{R} (-V_{C1} + v(t) - V_{C2}) \right) \right]$$

$$\frac{dx_1}{dt} = \frac{1}{C_1} x_2 - \frac{1}{RC_1} x_1 - \frac{1}{RC_1} x_3 + \frac{1}{RC_1} v(t)$$

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \end{bmatrix} = \begin{bmatrix} -1/RC_1 & 1/C_1 & -1/RC_1 \\ -1/L & 0 & 0 \\ -1/RC_2 & 0 & -1/RC_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1/RC_1 \\ 1/L \\ 1/RC_2 \end{bmatrix} v(t)$$

$$\xrightarrow{V_0(t)} \\ y(t) = V_{C2}$$

$$y = x_3$$

$$[y] = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

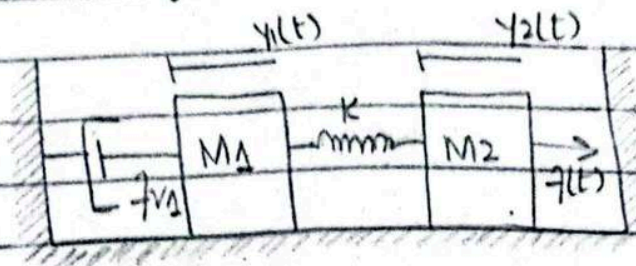
State Representation of Mechanical Systems

$\therefore$ is obtained from point of motion/displacement rather than energy storing elements.

∴ Take displacement rather and velocity at each point as state space variables.

→ if its a 2nd state equation

$$\left[\begin{array}{c} \text{num of} \\ \text{state variable} \\ \text{in equation} \end{array} \right] = \left[\begin{array}{c} \text{equal to the} \\ \text{order of that} \\ \text{equation} \end{array} \right]$$



$$\rightarrow M_1 \frac{d^2 y_1}{dt^2} + f_{v1} \frac{dy_1}{dt} + k(y_1 - y_2) = 0 \quad \text{--- (i)}$$

$$\rightarrow M_2 \frac{d^2 y_2}{dt^2} + k(y_2 - y_1) = f(t) \quad \text{--- (ii)}$$

$$y_1 = x_1, \quad \frac{dy_1}{dt} = x_2 \Rightarrow \frac{dx_1}{dt} = x_2 \Rightarrow \frac{dx^2}{dt} = \frac{dx_1}{dt}$$

$$y_2 = x_3, \quad \frac{dy_2}{dt} = x_4 \Rightarrow \frac{dx_3}{dt} = x_4 \Rightarrow \frac{dx_4}{dt} = \frac{d^2 y_2}{dt^2}$$

$$\frac{d^2 y_1}{dt^2} = \frac{-k}{M_1} y_1 + \frac{k}{M_1} y_2 - \frac{f_{v1}}{M_1} \frac{dy_1}{dt}$$

$$\frac{dx_2}{dt} = \frac{-k}{M_1} x_1 + \frac{k}{M_2} x_3 - \frac{f_{v1}}{M_1} x_2 \quad \text{--- *}$$

Similarly

$$\frac{d^2 y_2}{dt^2} = \frac{k}{M_2} y_1 - \frac{k}{M_2} y_2 + \frac{1}{M_2} f(t)$$

$$\frac{dx_4}{dt} = \frac{k}{M_2} x_1 - \frac{k}{M_2} x_3 + \frac{1}{M_2} f(t)$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \\ dx_4/dt \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -k/M_1 & -7v_1/M_1 & k/M_1 & 0 \\ 0 & 0 & 0 & 1 \\ k/M_2 & 0 & -k/M_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/M_2 \end{bmatrix} f(t)$$

$$y_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad \therefore y_1 = x_1$$

$$y_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad \therefore y_2 = x_3$$

State space representation from Transfer Fn

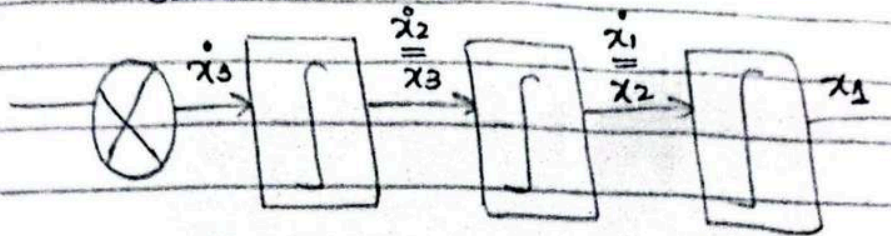
- ∴ The no. of state variables = the order of system
- ∴ Select set of state variables in such a way that each variable is derivative of prev variable.

$$\frac{Cs}{Rs} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

$$C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

∴ No. of integrators will be equal to no. of state variables

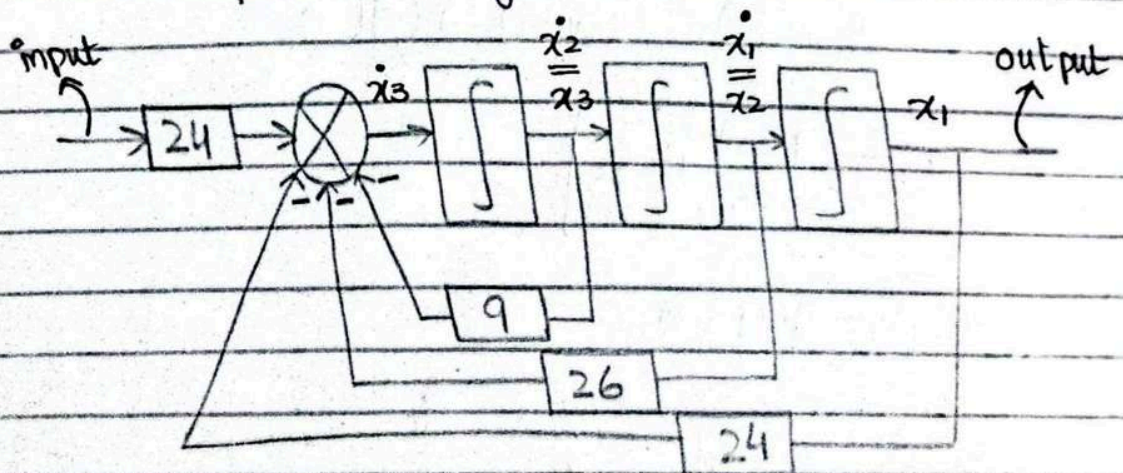


Now by matrix:

$$\frac{d^3c}{dt^3} + 9\frac{d^2c}{dt^2} + 26\frac{dc}{dt} + 24 = 24r$$

$$c = x_1, \frac{dc}{dt} = x_2, \frac{dx_2}{dt} = \frac{d^2c}{dt^2} = x_3$$

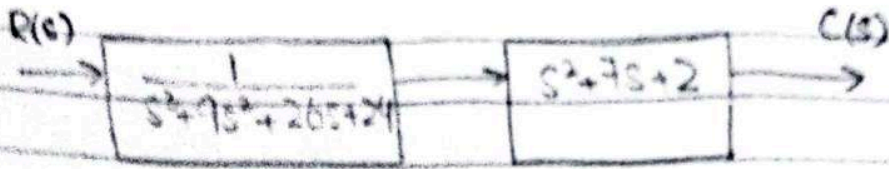
Now updated diagram will be



* transfer function is defined for having one input and one output



$$\frac{C(s)}{R(s)} = \frac{s^2 + 7s + 2}{s^3 + 9s^2 + 26s + 24}$$



$$Y(s) = \frac{1}{s^3 + 9s^2 + 26s + 24} \cdot (R(s))$$

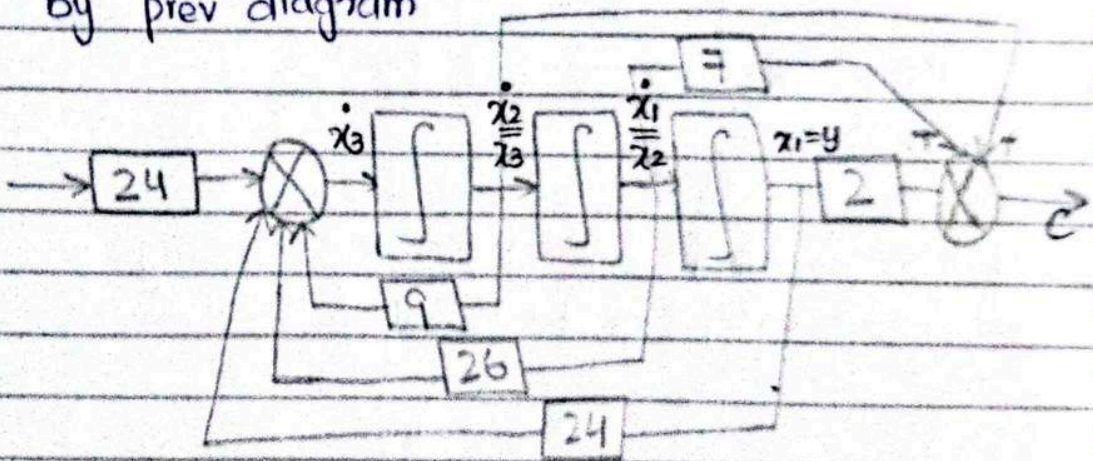
$$y = x_2 \quad ; \quad \frac{dy_1}{dx} = x_2 \quad ; \quad \frac{d^2y}{dx^2} = x_3$$

$$c(t) = \frac{d^2 y(t)}{dt^2} + 7 \frac{dy(t)}{dt} + 2y(t)$$

↳ From $C(s) = (s^2 + 7s + 2)(Y(s))$

$$C = \begin{bmatrix} 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

by prev diagram



Converting state space Model into T.F :

$$\dot{X} = AX + BU$$

$$sX(s) = AX(s) + BU(s)$$

$$Y = CX + DU$$

$$Y(s) = CX(s) + DU(s)$$

$$sX(s) - AX(s) = BU(s)$$

$$X(s)[sI - A] = BU(s)$$

$$X(s) = [sI - A]^{-1} BU(s)$$

$$Y(s) = C[sI - A]^{-1} BU(s) + DU(s)$$

$$= [C[sI - A]^{-1} B + D] U(s)$$

$$\frac{Y(s)}{U(s)} = C[sI - A]^{-1} B + D$$

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow sI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 2 & s+3 \end{bmatrix}$$

$$(sI-A)^{-1} = \frac{\text{Adj}(sI-A)}{\det(sI-A)}$$

$$\det(sI-A) = s(s(s+3)+2)+1[0+1] \\ = s^3+3s^2+2s+1$$

$$\text{Adj} A = [\text{Matrix of co-factors}]_T$$

$$\text{Matrix of co-factors} = \begin{bmatrix} s(s+3)+2 & -1 & -s \\ s+3 & s^2+3s & -(2s+1) \\ 1 & s & s^2 \end{bmatrix}$$

$$\text{Adj}(sI-A) = \begin{bmatrix} s^2+3s+2 & s+3 & 1 \\ -1 & s^2+3s & s \\ -s & -(2s+1) & s^2 \end{bmatrix}$$

$$T.F = C(sI-A)^{-1}B+D$$

$$= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \frac{1}{\Delta} \begin{bmatrix} s^2+3s+2 & s+3 & 1 \\ -1 & s^2+3s & s \\ -s & -(2s+1) & s^2 \end{bmatrix} \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\Delta} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 10(s^2+3s+2) \\ -10 \\ -10s \end{bmatrix}$$

$$= \frac{10(s^2+3s+2)}{s^3+3s^2+2s+1}$$

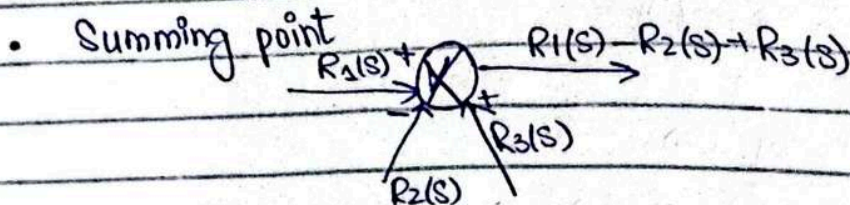
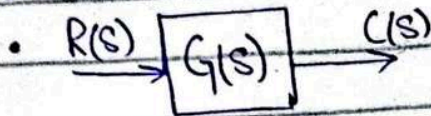


Chapter no 5

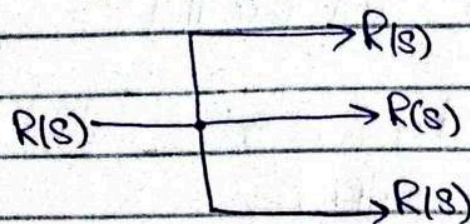
BLOCK DIAGRAM REDUCTION

Components of a block diagram

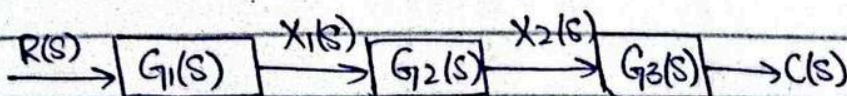
C(s)



- Pick off point / Branch point



→ CASCADE System:



$$X_1(s) = G_1(s) R(s)$$

$$X_2(s) = G_2(s) X_1(s) = G_2(s) G_1(s) R(s)$$

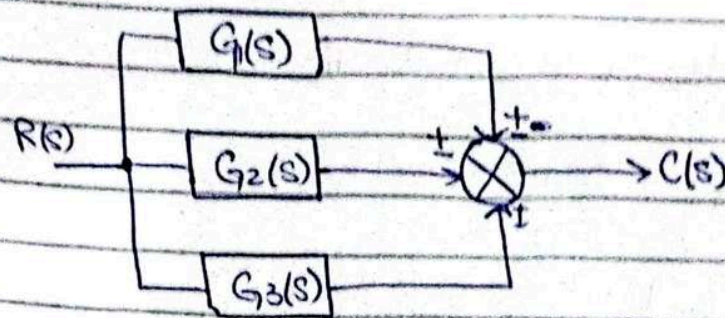
$$C(s) = G_3(s) X_2(s)$$

$$= G_3(s) G_2(s) G_1(s) R(s)$$

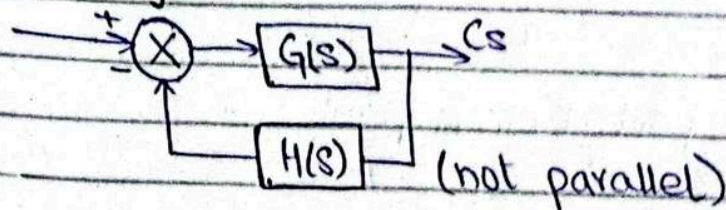
$$\frac{C(s)}{R(s)} = G_3(s) G_2(s) G_1(s) \quad (\text{Transfer function})$$

Parallel sub-systems

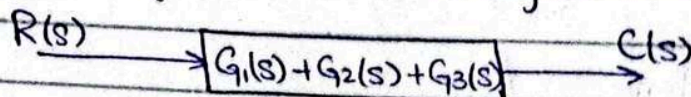
i/p and o/p at the same point



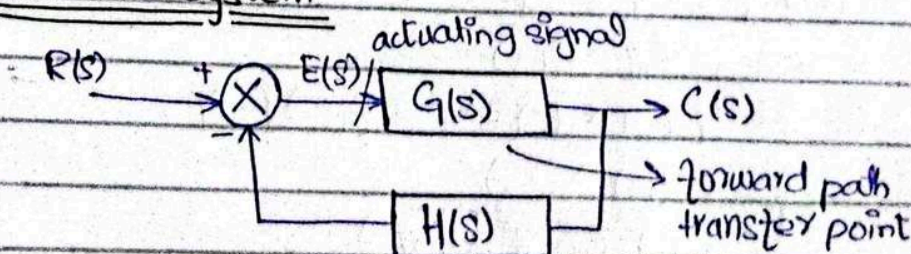
feedback system



$$C(s) = G_1(s)R(s) + G_2(s)R(s) + G_3(s)R(s)$$
$$= [G_1(s) + G_2(s) + G_3(s)]R(s)$$



Feedback system



$\therefore$ actuator - signal at summing point

$$C(s) = G(s) E(s) \quad (i)$$

$$E(s) = R(s) - B(s)$$

$$B(s) = H(s)G(s)$$

$$E(s) = R(s) - H(s)G(s)$$

$$\Rightarrow (i) \quad C(s) = G(s) \cdot [R(s) - H(s)G(s)]$$

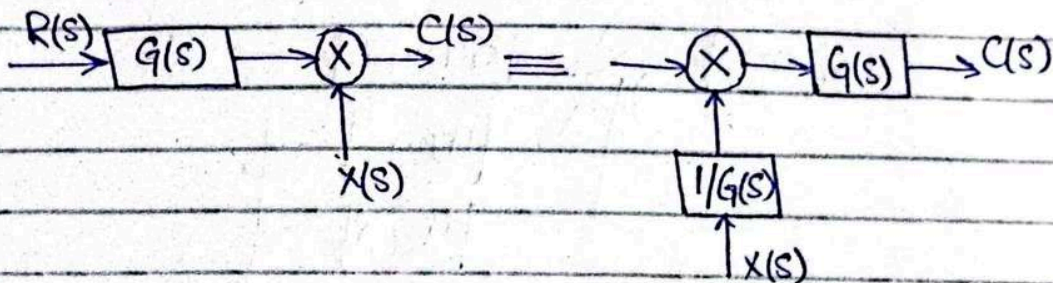
$$C(s) + G(s)H(s)C(s) = G(s)R(s)$$

$$C(s)[1 + G(s)H(s)] = G(s)R(s)$$

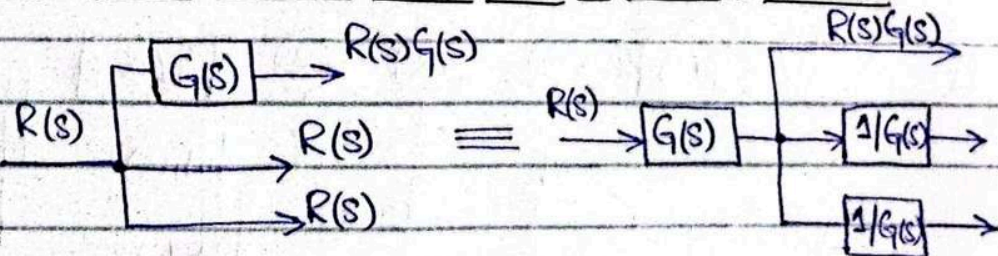
$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \quad \begin{array}{l} \text{(closed loop transfer} \\ \text{function)} \end{array}$$

open loop transfer function

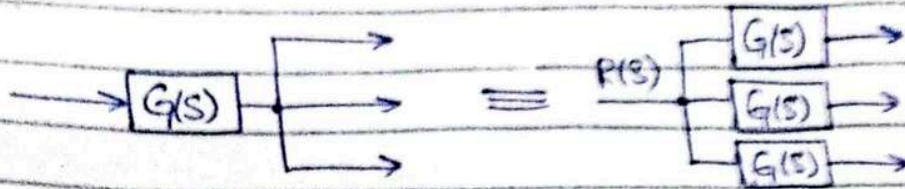
→ Moving a transfer function to the left of summing point



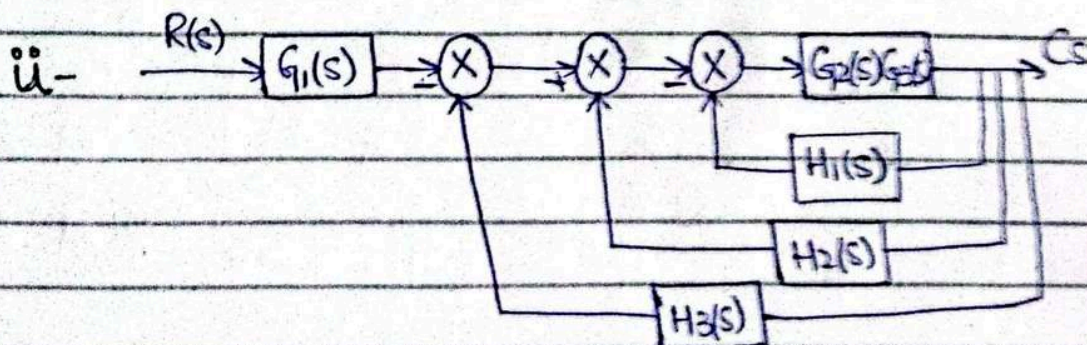
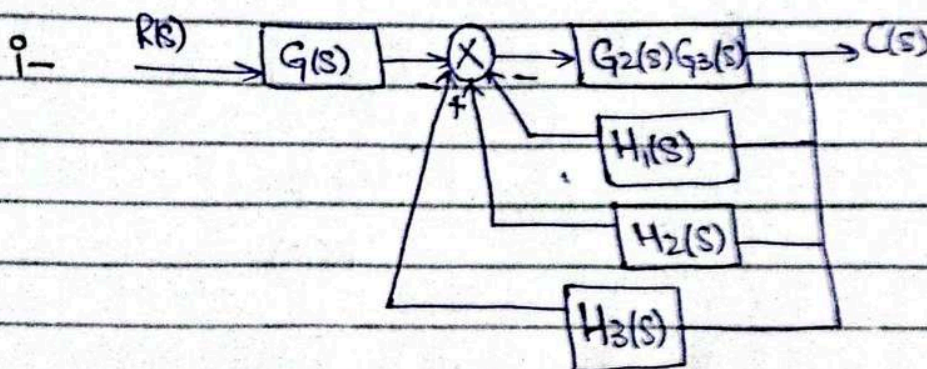
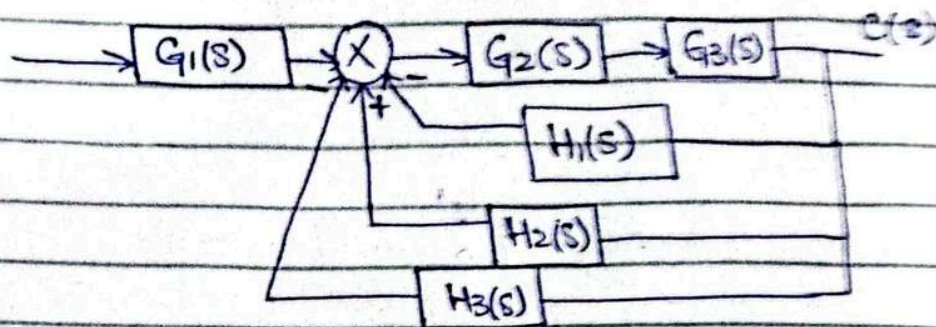
→ Moving a T.F to the left of pick off point

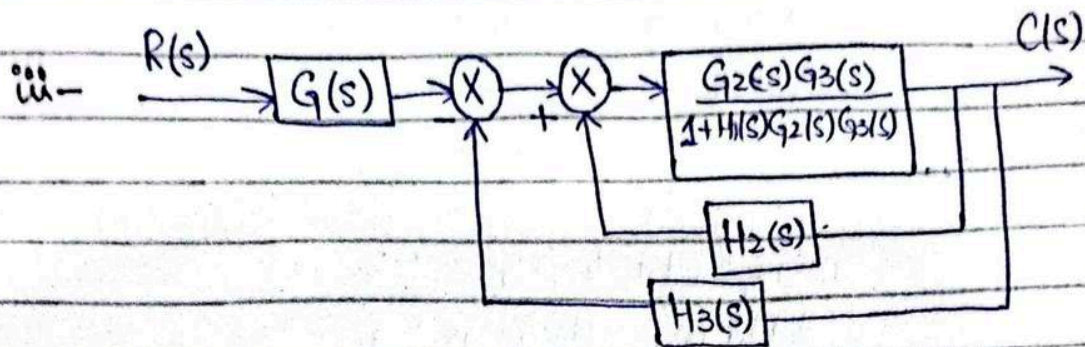


→ Moving a TF to the right of pick off point:

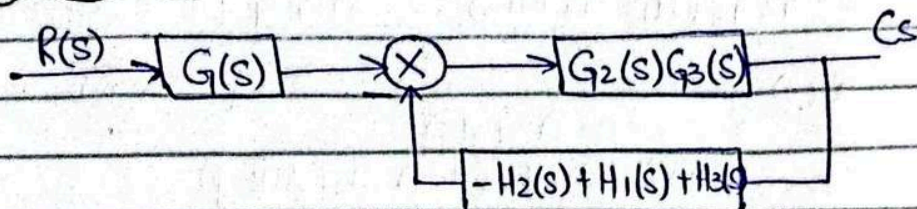


Example #
Method I





Method II

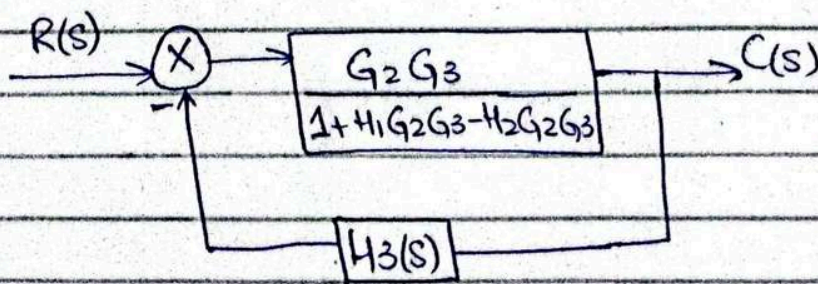


$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3}{1 + G_2 G_3 (H_1 - H_2 + H_3)}$$

iv-

$$\frac{G_2 G_3 / (1 + H_1 G_2 G_3)}{1 - \left(\frac{G_2 G_3}{1 + H_1 G_2 G_3} \right) \times H_2} = \frac{G_2 G_3 / (1 + H_1 G_2 G_3)}{1 + H_1 G_2 G_3 - G_2 G_3 H_2}$$

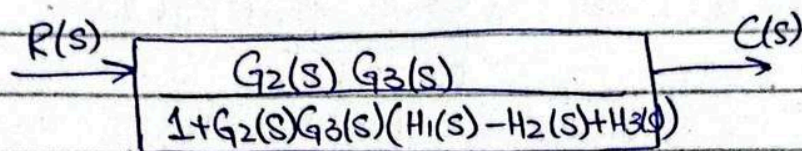
$$= \frac{G_2 G_3}{1 + H_1 G_2 G_3 - H_2 G_2 G_3}$$



$$V = \frac{G_2 G_3}{1 + H_1 G_2 G_3 - H_2 G_2 G_3} \times H_3$$

$$= \frac{G_2 G_3}{1 + H_1 G_2 G_3 - H_2 G_2 G_3 + G_2 G_3 H_3}$$

$$= \frac{G_2 G_3}{1 + G_2 G_3 (H_1 - H_2 + H_3)} = \frac{G_2 G_3}{1 + G_2 G_3 (H_1 - H_2 + H_3)}$$

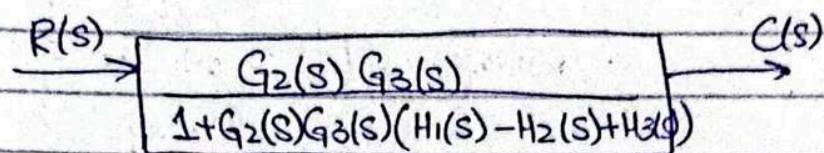


$$\frac{C(s)}{R(s)} = \frac{G_2(s) G_3(s)}{1 + G_2(s) G_3(s) (H_1(s) - H_2(s) + H_3(s))}$$

$$V = \frac{G_2 G_3}{1 + H_1 G_2 G_3 - H_2 G_2 G_3} \times H_3$$

$$= \frac{G_2 G_3}{1 + H_1 G_2 G_3 - H_2 G_2 G_3 + G_2 G_3 H_3}$$

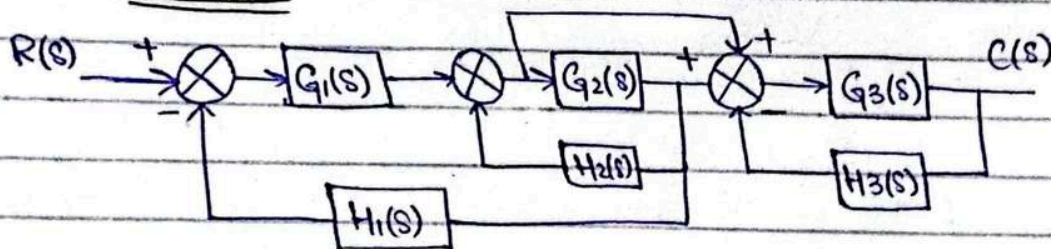
$$= \frac{G_2 G_3}{1 + G_2 G_3 (H_1 - H_2 + H_3)} = \frac{G_2 G_3}{1 + G_2 G_3 (H_1 - H_2 + H_3)}$$



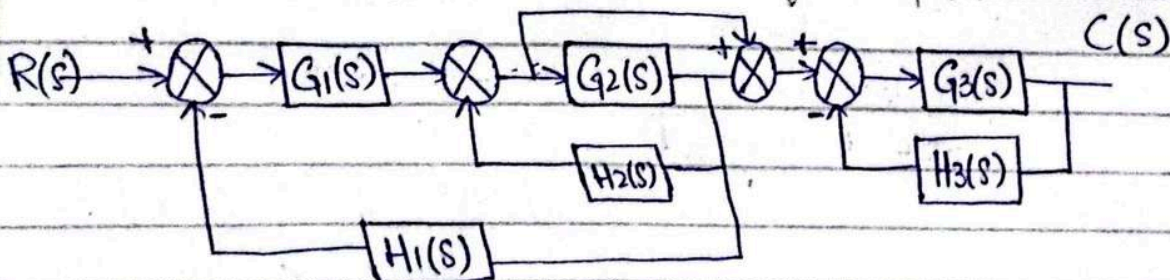
$$\frac{C(s)}{R(s)} = \frac{G_2(s) G_3(s)}{1 + G_2(s) G_3(s) (H_1(s) - H_2(s) + H_3(s))}$$

(12)

Example:



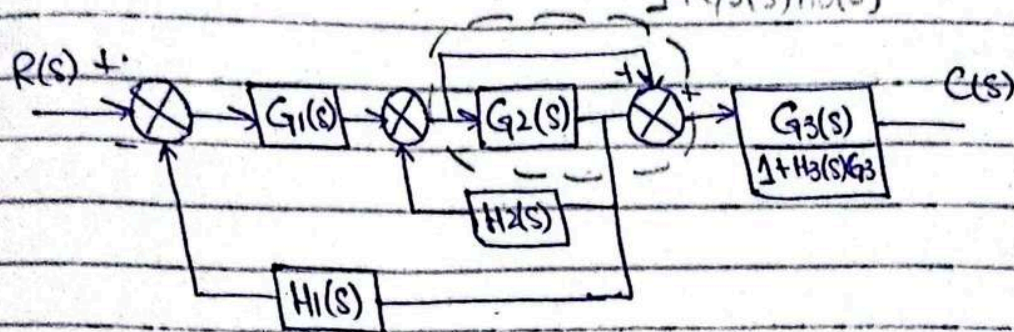
$\therefore$ we will put an adder before $G_3(s)$



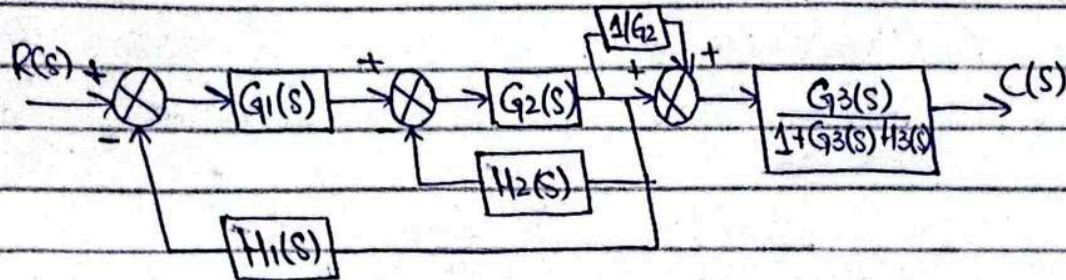
Now, $G_3(s)$ and $H_3(s)$ are in -ve Loop

We can solve them

So, it will be $= \frac{G_3(s)}{1 + G_3(s)H_3(s)}$

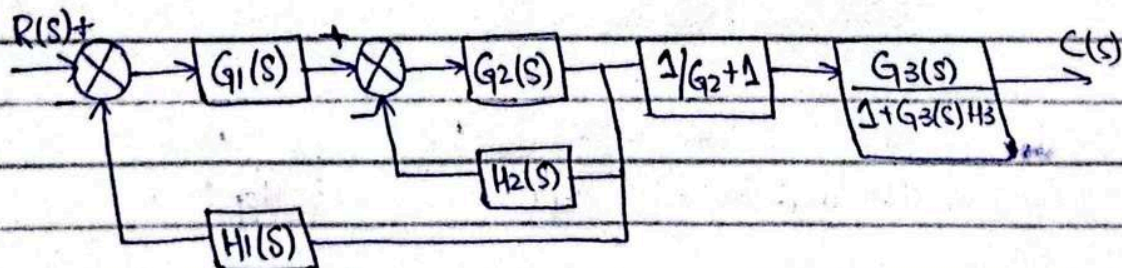


Now we can solve the closed feedback Loop



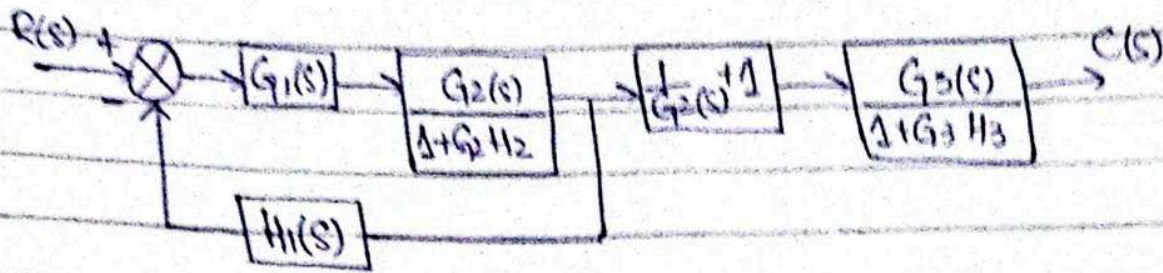
Now we will solve $G_2(s)$ and $\frac{1}{G_2(s)}$

∴ We cannot interchange summing and pick off point



Now, G_2 and H_2 are in -ve feedback Loop So,

$$= \frac{G_2(s)}{1 + G_2(s)H_2(s)}$$



Now $G_1(s)$ and $\frac{G_2(s)}{1+G_2H_2}$ are in series

then they're in negative feedback with $H_1(s)$ and then they're in series with next two blocks.

Solving the expression:

$$= \frac{G_1(s) \left(\frac{G_2(s)}{1+G_2H_2} \right)}{\quad} \quad \therefore \text{in series}$$

$$= \frac{G_1G_2}{1+G_2H_2} \quad (\text{negative feedback with } H_1(s))$$

$$= \frac{G_1G_2 / (1+G_2H_2)}{1 + \frac{G_1G_2}{1+G_2H_2} \cdot H_1} = \frac{G_1G_2 / (1+G_2H_2)}{1 + G_2H_2 + G_1G_2H_2}$$

$$= \frac{G_1G_2}{1+G_2H_2+G_1G_2H_2}$$

Now solving all three blocks in series

$$= \frac{G_1G_2}{1+G_2H_2+G_1G_2H_2} \cdot \frac{1+G_2}{G_2} \cdot \frac{G_3}{1+G_3H_3}$$

$$= \frac{G_1G_3(1+G_2)}{(1+G_2H_2+G_1G_2H_2)(1+G_3H_3)} = \frac{C(s)}{R(s)}$$

Series $\rightarrow$ Multiply

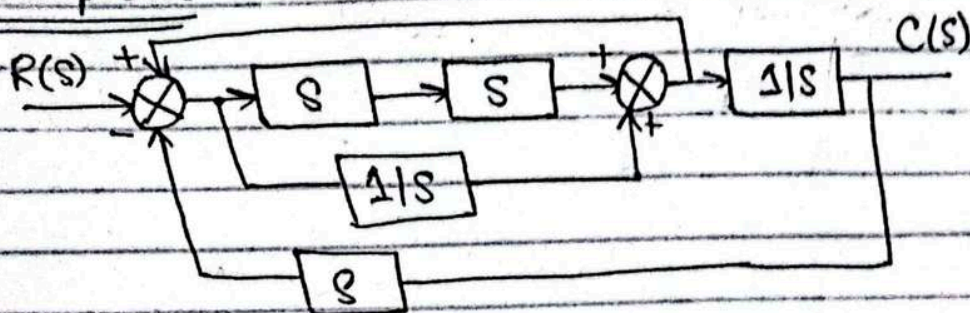
Parallel $\rightarrow$ Addition

Negative feedback $\rightarrow$

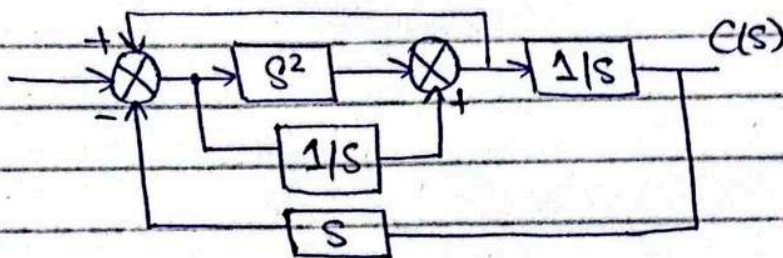
Unity close loop $\rightarrow \frac{G_1(s)}{1+G_1(s)}$

$$\frac{G_1(s)}{1+G_1(s)(H_1(s))}$$

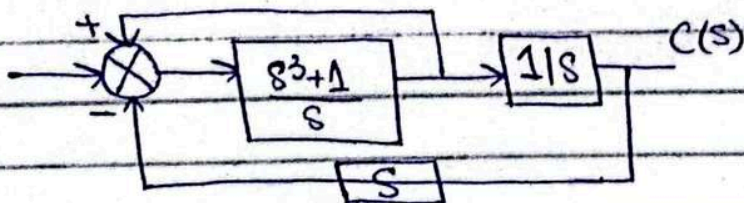
Example 2:



$$= (s)(s) = s^2 \text{ (series)}$$

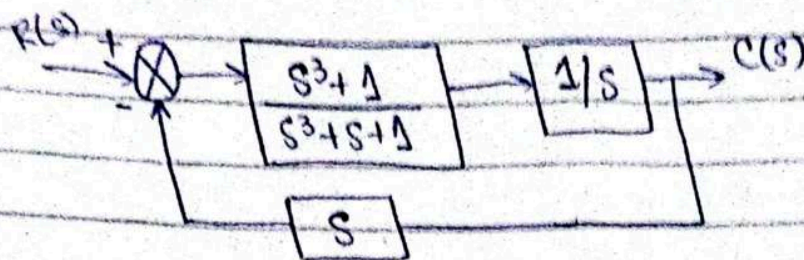


$$= s^2 + \frac{1}{s} = \frac{s^3 + 1}{s} \text{ (parallel)}$$

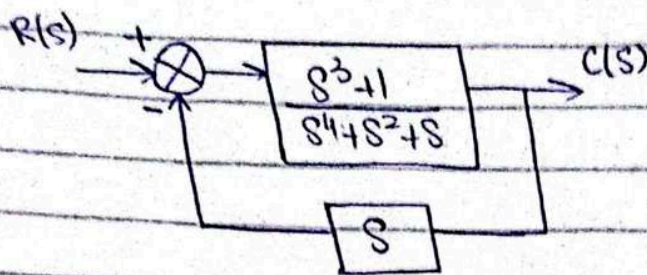


Solving closed loop

$$= \frac{s^3 + 1/s}{1 + s^3 + 1/s} = \frac{s^3 + 1/s}{1 + s^3 + s/s} = \frac{s^3 + 1}{s^3 + s + 1}$$



$$= \left(\frac{s^3 + 1}{s^3 + s + 1} \right) \left(\frac{1}{s} \right) = \frac{s^3 + 1}{s^4 + s^2 + s} \quad (\text{series})$$



$$= \frac{s^3 + 1}{s^4 + s^2 + s} = \frac{s^3 + 1}{s^4 + s^2 + s} \cdot \frac{s}{s} = \frac{s^3 + 1}{s^4 + s^2 + s} \cdot \frac{s}{s} = \frac{s^3 + 1}{s^4 + s^2 + s} \cdot \frac{s}{s}$$

$$= \frac{s^3 + 1}{2s^4 + s^3 + 2s}$$

So,

$$\frac{R(s)}{C(s)} = \frac{s^3 + 1}{2s^4 + s^3 + 2s}$$

Chapter no 04

Time Analysis of first order system

$$\frac{C(s)}{R(s)} = G(s) = \frac{k}{s+a} \quad \begin{array}{c|c} \times & \times \\ \hline a & \end{array}$$

Total Response = Forced Response + Natural Response

Step response:

$$C(s) = \frac{k}{s(s+a)}$$

$$= \frac{k/a}{s} - \frac{k/a}{s+a}$$

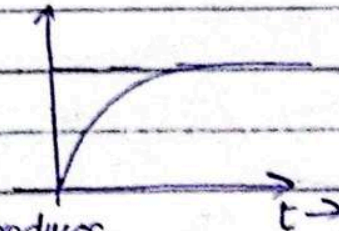
forced natural

∴ the output due to poles is natural

∴ the output due to poles of input is forced

∴ In LTI, complex poles always exist in conjugate pairs.

$$c(t) = \frac{k}{a} (1 - e^{-at})$$



Every real pole in s-plane introduces an exp term in final response. the rate of decay or fall depends on the location pole.

$$G(s) = \frac{s+2}{s+5} ; C(s) = \frac{s+2}{s(s+5)}$$

$$= \frac{A}{s} + \frac{B}{s+5}$$

$$= \frac{2/5}{s} - \frac{3/5}{s+5}$$

$$c(t) = \frac{2}{5}u(t) - \frac{3}{5}e^{-5t}u(t)$$

(15)

Transient Response of first order system:

1- Time Constant:-

Time for step response to reach 63% of its final value.

OR

Time for e^{-at} to decay to 37% of its initial value.

2- Rise Time:-

Time for the waveform to go from 0.1-0.9 (10-90%) of its final value

3- Settling Time:-

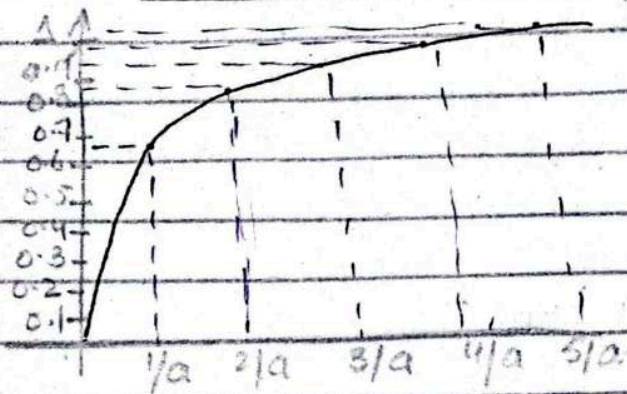
Time for the response to reach and stay within 2% of its final value.

$$G(s) = \frac{k}{s+a} \quad ; \quad c(t) = \frac{k}{a} (1 - e^{-at})$$

Assume $k=a$

$$G(s) = \frac{a}{s+a} \quad ; \quad c(s) = \frac{a}{s(s+a)}$$

$$c(t) = (1 - e^{-at})u(t)$$



$$\begin{aligned} \rightarrow c(t) \text{ at } t=1/a \\ c(t) &= 1 - e^{-a(1/a)} \\ &= 0.623 \end{aligned}$$

$$\begin{aligned} \rightarrow t=2/a \\ &= 1 - e^{-a(2/a)} = 0.864 \end{aligned}$$

$$\begin{aligned} \rightarrow t=3/a \\ &= 0.950 \end{aligned}$$

$$\begin{aligned} \text{1st order system} &= \text{Time constant} \\ &= 1/a \end{aligned}$$

$\therefore a$ is the location of pole

$$\begin{aligned} \rightarrow t=4/a \\ &= 0.981 \end{aligned}$$

$$\begin{aligned} \rightarrow t=5/a \\ &= 0.993 \end{aligned}$$

$\hookrightarrow$ For Rise time:

$$T_r = t_2 - t_1$$

$$\hookrightarrow c(t) = 1 - e^{-at}$$

$$0.1 = 1 - e^{-at}$$

$$e^{-at} = 1 - 0.1$$

$$-at = \ln(1 - 0.1)$$

$$-at = -0.105$$

$$t_1 = \frac{0.105}{a}$$

$$\rightarrow c(t) = 1 - e^{-at}$$

$$0.9 = 1 - e^{-at}$$

$$e^{-at} = 1 - 0.9$$

$$\ln(e^{-at}) = \ln(1 - 0.9)$$

$$-at = -2.302$$

$$t_2 = \frac{2.302}{a}$$

$$T_r = t_2 - t_1 = \frac{2.302}{a} - \frac{0.105}{a} = \frac{2.2}{a}$$

→ For settling time:

$$c(t) = 1 - e^{-at}$$

$$0.98 = 1 - e^{-at}$$

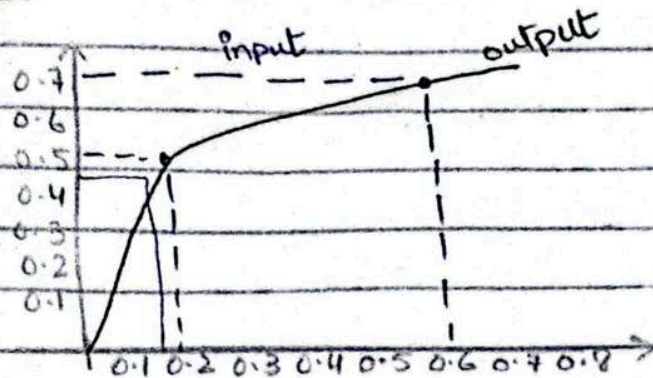
$$e^{-at} = 1 - 0.98$$

$$-at = -3.91$$

$$t = \frac{3.91}{a} \approx \frac{4}{a}$$

∴ It will take 4 time constants to reach the steady state.

Example:



$$\text{Final value} = 0.72$$

$$0.63 \times 0.72 = 0.45$$

$$\text{Time constant} = \frac{1}{a} = 0.15 \Rightarrow a = 6.66$$

$$c(t) = \frac{k}{a} (1 - e^{-at})$$

$$\frac{k}{a} = \text{final value}$$

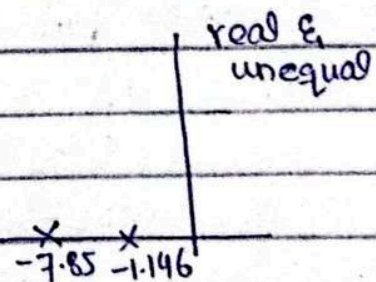
$$k = 0.72 \times a = 0.72 \times 6.6 = 4.75$$

$$G(s) = \frac{4.75}{s + 6.6}$$

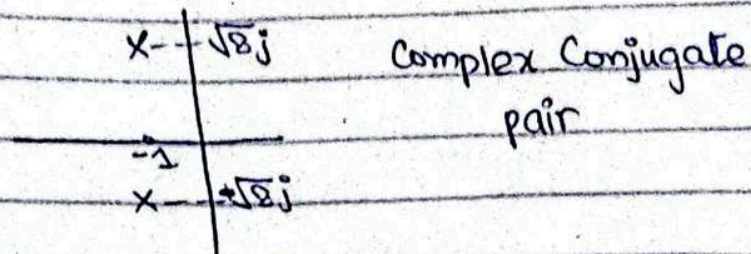
Transient Response of second order system

$$G_1(s) = \frac{9}{s^2 + 9s + 9}$$

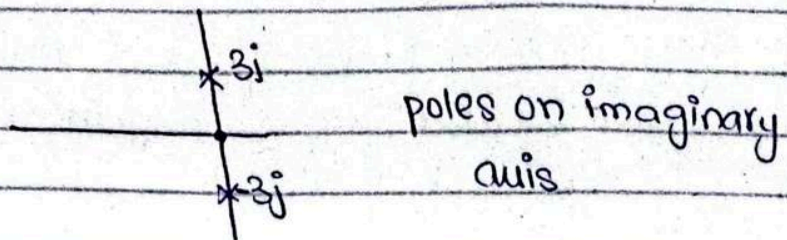
$$= \frac{9}{(s + 7.854)(s + 1.146)}$$



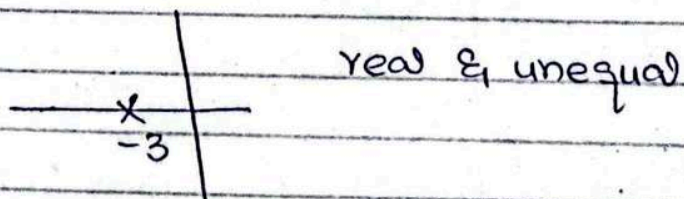
$$\rightarrow G_2(s) = \frac{9}{s^2+2s+9} = \frac{9}{(s+1+j\sqrt{8}) (s+1-j\sqrt{8})}$$



$$\rightarrow G_3(s) = \frac{9}{s^2+9} = \frac{9}{(s+3j)(s-3j)}$$



$$\rightarrow G_4(s) = \frac{9}{s^2+6s+9} = \frac{9}{(s+3)^2}$$



$\Rightarrow$ By partial fraction expansion

$$c_1(t) = 1 + 0.171e^{-7.854t} - 1.171e^{-1.146t}$$

$$c_2(t) = 1 - 1.06e^{-t} \cos(\sqrt{8}t - 19.47^\circ)$$

$$c_3(t) = 1 - 3e^{-3t} \cos 3t$$

$$c_4(t) = 1 - 3te^{-3t} - e^{-3t}$$

ω_n = natural frequency

ζ = damping rate

$\therefore$ Damping is actually the rate of decay and fall.

$\therefore$ Damping is actually the rate of decay and fall.

Second Order System

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ω_n = Natural frequency

ζ = Damping ratio

$\zeta = 0 \Rightarrow$ undamped system

$0 < \zeta < 1 \Rightarrow$ underdamped system

$\zeta = 1 \Rightarrow$ critically damped

$\zeta > 1 \Rightarrow$ overdamped system

$$\rightarrow \frac{C(s)}{R(s)} = G(s) = \frac{9}{s^2 + 9s + 9}$$

$$R(s) = \frac{1}{s}$$

$$r(t) = u(t)$$

$$C(s) = \frac{9}{s(s+7.854)(s+1.146)}$$

$$\rightarrow c(t) = 1 + 0.17e^{-7.854t} - 1.171e^{-1.146t}$$

$\therefore$ consider ω_n^2

$$\omega_n^2 = 9 \Rightarrow \omega_n = 3$$

$$2\zeta\omega_n = 9$$

$$\zeta = \frac{9}{6} = 1.5 > 1 \text{ (overdamped)}$$

from denominator

$$\rightarrow G_2(s) = \frac{9}{s^2 + 2s + 9}$$

$$\omega_n^2 = 9 \Rightarrow \omega_n = 3$$

$$2\zeta\omega_n = 2 \Rightarrow \zeta = \frac{2}{6} = \frac{1}{3}$$

$$0 < \zeta < 1 \text{ (underdamped)}$$

$$c(t) = 1 - 1.06e^{-t} \cos(\sqrt{8}t - 19.47)$$

$$\rightarrow G_3(s) = \frac{9}{s^2 + 9}$$

$$\omega_n = 3 ; 2\zeta\omega_n = 0 ; \zeta = 0 \text{ (undamped)}$$

$$c(t) = 1 - \cos 3t$$

$$\rightarrow G_4(s) = \frac{9}{(s+3)^2} = \frac{9}{s^2 + 6s + 9}$$

$$2\zeta\omega_n = 6 ; \zeta = 1 ; \omega_n = 3$$

(critically damped)

$$c(t) = 1 - 3te^{-3t} - e^{-3t}$$

Transient Response of Underdamped 2nd order System

1- Rise times

Time to reach 10-90% of final value.

2- Peak Time:

Time to reach 1st peak of the response

3- Maximum Overshoot

The amount that the waveform overshoots the steady state or final value at the peak time expressed as a percentage of steady state value

$$\% OS = \frac{C_{max} - C_{final}}{C_{final}} \times 100\%$$

4- Settling time

The time constants they take to reach steady state.

Response of second order system

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$A = 1 ; B = -1 ; C = -2\zeta\omega_n$$

$$C(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2 + \omega_n^2\zeta^2 - \omega_n^2\zeta^2}$$

$$= \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} \quad \because \omega_n^2(1 - \zeta^2) = \omega_n^2$$

$$= \frac{1}{s} - \left[\frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2} + \frac{\zeta\omega_n(\sqrt{1 - \zeta^2})}{\sqrt{1 - \zeta^2}[(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2]} \right]$$

$$c(t) = 1 - \left[e^{-\zeta\omega_n t} \cos(\omega_n\sqrt{1 - \zeta^2}t) + \frac{\zeta}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n\sqrt{1 - \zeta^2}t) \right]$$

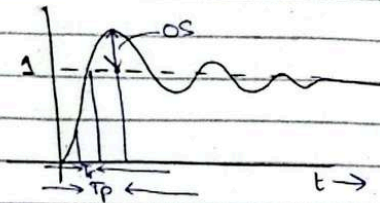
$$= \frac{1 - e^{-\zeta \omega_n t}}{\sqrt{1 - \zeta^2}} \left[\sqrt{1 - \zeta^2} \cos(\omega_n \sqrt{1 - \zeta^2} t) + \zeta \sin(\omega_n \sqrt{1 - \zeta^2} t) \right]$$

$$\therefore \cos A \cos B + \sin A \sin B = \cos(A - B)$$

$$c(t) = \frac{1 - e^{-\zeta \omega_n t}}{\sqrt{1 - \zeta^2}} \left[\cos(\omega_n \sqrt{1 - \zeta^2} t - \phi) \right]$$

$$\boxed{\phi = \tan^{-1} \frac{\zeta}{\sqrt{1 - \zeta^2}}} \rightarrow \text{only remember this}$$

Now will use this expression to find peak time and overshoot.



Peak Time T_p

$$\boxed{T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}}$$

we can also derive this formula but I'm not gonna do it here.

Max Overshoot

$$c(t) = \frac{1 - e^{-\zeta \omega_n t}}{\sqrt{1 - \zeta^2}} \left(\cos \omega_n \sqrt{1 - \zeta^2} t + \zeta \sin \omega_n \sqrt{1 - \zeta^2} t \right)$$

$$C_{max} = C(T_p) \therefore T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

$$c(t) = \frac{1 - e^{-\frac{\zeta \omega_n t}{\sqrt{1 - \zeta^2}}}}{\sqrt{1 - \zeta^2}} \left(\cos \omega_n \sqrt{1 - \zeta^2} t + \zeta \sin \omega_n \sqrt{1 - \zeta^2} t \right)$$

$$C_{max} = 1 + e^{-\frac{\zeta \pi}{\sqrt{1 - \zeta^2}}} \rightarrow \text{remember this only}$$

$$\boxed{\%OS = e^{-\frac{\zeta \pi}{\sqrt{1 - \zeta^2}}} \times 100}$$

Location of poles of 2nd order system in terms of ζ and ω_n

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$$s^2 + 2\zeta \omega_n s + \omega_n^2 = 0$$

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

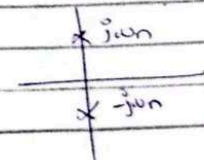
$$s = \frac{-2\zeta \omega_n \pm \sqrt{4\zeta^2 \omega_n^2 - 4\omega_n^2}}{2}$$

$$= \frac{-2\zeta \omega_n \pm 2\omega_n \sqrt{\zeta^2 - 1}}{2}$$

$$= \frac{-2\zeta \omega_n \pm 2\omega_n \sqrt{\zeta^2 - 1}}{2}$$

$\rightarrow$ Undamped Condition

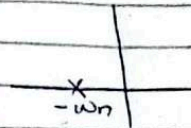
$$\zeta = 0 ; s = \pm j\omega_n$$



→ Critically damped condition

$$\zeta = 1$$

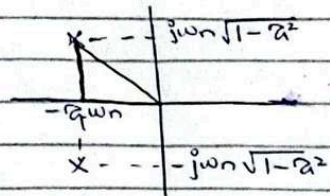
$$s = -\omega_n \pm 0$$



→ Underdamped condition

$$0 < \zeta < 1$$

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$



$$\begin{aligned} r = \text{root} &= \sqrt{\zeta^2\omega_n^2 + \omega_n^2(1-\zeta^2)} \\ &= \sqrt{\zeta^2\omega_n^2 + \omega_n^2(1-\zeta^2)} \\ &= \sqrt{\zeta^2\omega_n^2 + \omega_n^2 - \omega_n^2\zeta^2} = \omega_n \end{aligned}$$

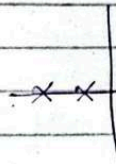
$$\cos\theta = \frac{\zeta\omega_n}{\omega_n} \Rightarrow \theta = \cos^{-1}(\zeta)$$

→ Overdamped condition

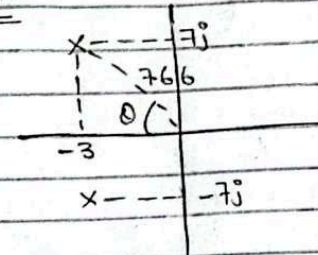
$$\zeta > 1$$

$$s = -\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}$$

$$s = -\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}$$



Question



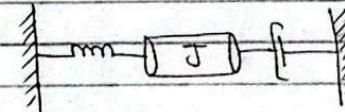
Find $\zeta, \omega_n, T_p, \%OS, T_s = p$

$$\begin{aligned} \theta &= \tan^{-1} \frac{7}{3} ; \omega_n = \sqrt{7^2 + 3^2} ; \zeta = \cos\theta \\ &= \sqrt{49+9} ; \zeta = 0.394 \\ &= \sqrt{58} \\ &= 7.616 \end{aligned}$$

$$T_p = \frac{\pi}{\omega_n} ; T_s = \frac{4}{\zeta\omega_n} ; OS = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$T_p = 0.449 \text{ sec} ; T_s = 1.33 \text{ sec} ; OS = 26\%$$

Question



$$k = 5 \text{ Nm/rad}$$

Find J and D to yield 20% OS and $T_s = 2 \text{ sec}$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d} ; T_s = \frac{4}{2\omega_n}$$

$$\%OS = e^{\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$Qs(Js^2 + Ds + K) = T_s$$

$$\frac{Qs}{T_s} = \frac{1}{Js^2 + Ds + K}$$

$$\frac{J}{J} \times \frac{1}{Js^2 + Ds + K} = \frac{1/J}{s^2 + \frac{D}{J}s + \frac{K}{J}}$$

$$\omega_n^2 = \frac{K}{J} ; 2\zeta\omega_n = \frac{D}{J}$$

$$\omega_n = \sqrt{\frac{K}{J}} ; \zeta = \frac{D}{J} \cdot \frac{1}{2\omega_n}$$

$$0.2 = e^{\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$\zeta = -\ln(0.2/100)$$

$$\sqrt{\pi^2 + \ln^2(100/0.2)}$$

$$20\% OS \rightarrow \zeta = 0.456$$

$$T_c = 2 \text{ sec}$$

$$T_s = \frac{4}{2\omega_n} \Rightarrow 2\omega_n = \frac{4}{2} \Rightarrow \omega_n = \frac{2}{2} = 1$$

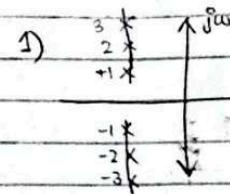
Chapter # 4

Relationship b/w ' ζ ' and 'overshoot'

* Every time corresponds to ζ and overshoot
 $\rightarrow$ inverse relation (b/w ζ and overshoot)

overshoot $\uparrow$ $\zeta \downarrow$

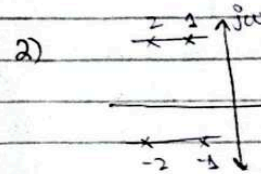
$\rightarrow$ direct relation (b/w rise time and ζ)
 $\zeta \uparrow$ rise time $\uparrow$



$$(\text{peak time})_{\max} = 1 ; (\text{peak time})_{\min} = 3$$

$$(\text{overshoot})_{\max} = 3 ; (\text{overshoot})_{\min} = 1$$

$\therefore$ all poles on vertical lines have same 'settling time'

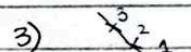


peak time $\rightarrow$ same

$$(\text{settling time})_{\max} = 1$$

$$(\text{overshoot})_{\max} = 1$$

$\therefore$ poles closer to origin have high settling time



(ζ and OS) $\rightarrow$ same

(settling & T_p) $\rightarrow$ different

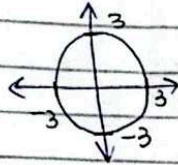
$$(\text{settling time})_{\max} = 1$$

$$(\text{peak time})_{\max} = 3$$

- * every real pole produces an exponential
- * if pole is far from origin, we neglect it

S-Plane

- Circle of 3 in s-plane



Question

if $z = 0.5$ & $\omega_n = 2$

Find transfer fn

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{4}{s^2 + 2s + 4}$$

- if peaks are same $\rightarrow \zeta$ and OS are same
- * small $\omega_n \rightarrow$ less oscillations
- * high $\omega_n \rightarrow$ more oscillations

• $\Rightarrow$ combination of three first order system

• $\Rightarrow$ pole-zero plot of third order system

Chapter # 6

Stability

Requirements of the design of control systems

- Transient Response
- Stability
- Steady state error

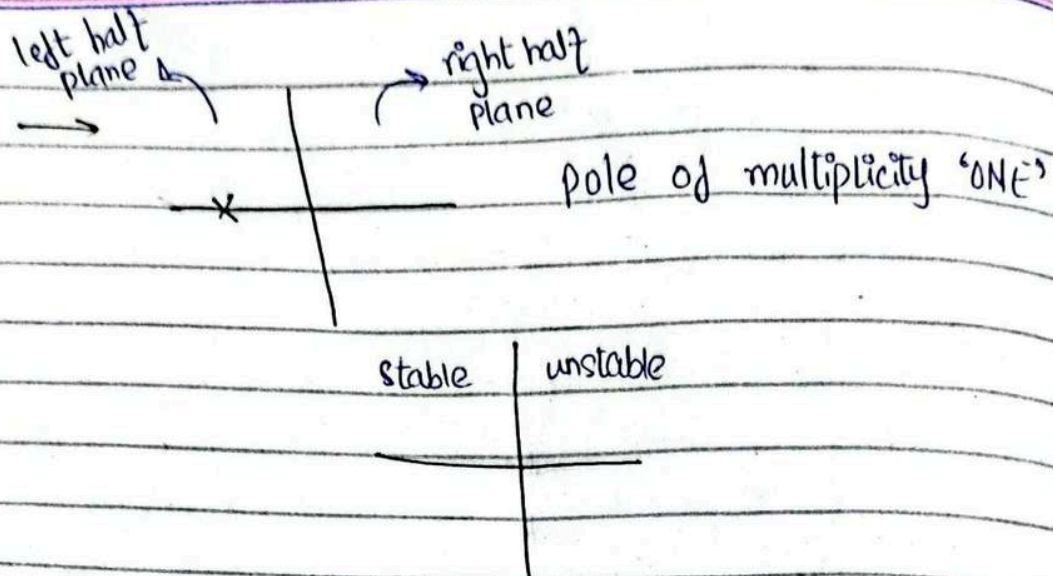
* System should be stable

- BIBO $\rightarrow$ stability in terms of total response
- BIUO $\rightarrow$ unstable

$\rightarrow$ using natural response

- if 't' in natural response approaches zero, it is a stable system
- if 't' approaches ∞ - unstable system
- if natural response neither decays nor grows but remain constant / oscillates, so it is a marginally stable system

$\rightarrow$ oscillatory response (Marginally stable system)



SUFFICIENT CONDITIONS FOR INSTABILITY

- ① Signs of denominator's coefficient are not same

Example:

$$\begin{array}{c} 3 \\ s^3 - 2s^2 + 5s + 3 \end{array}$$

(+1, -2, +5, +3) → not same

⇒ $1s^3 - 2s^2 + 5s + 3 = 0$ → characteristics Equation

- ② If any power of 's' is missing

Example:

$$\begin{array}{c} 3 \\ s^3 - 2s + 3 \end{array}$$

s^2 is missing